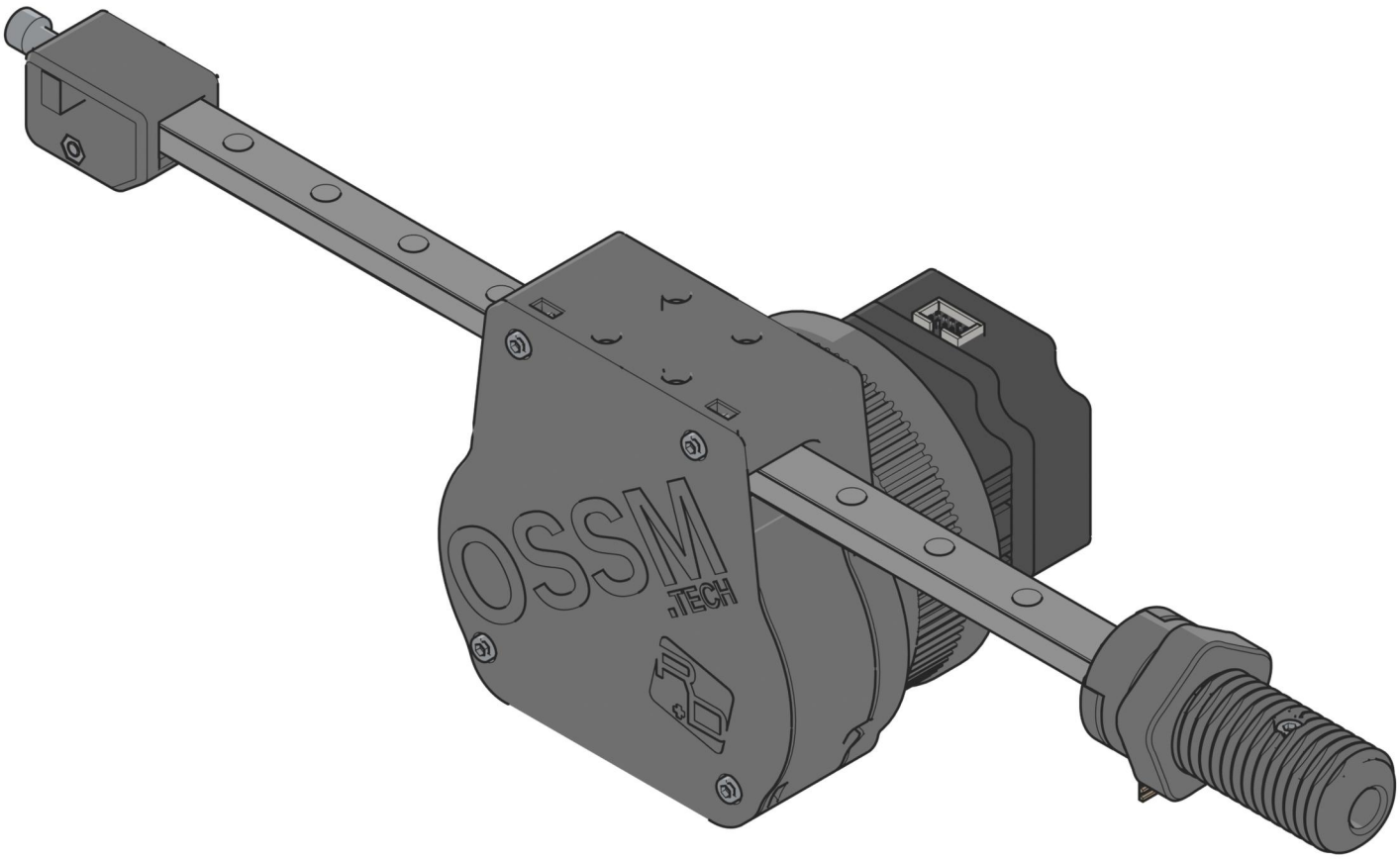
OSSM

Actuator Build Guide

Actuator · Version 1.1.13

RESEARCH + DESIRE



Support
support@researchanddesire.com

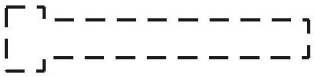
Assembly instructions
ohai.researchanddesire.com/ossm

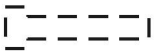
Actual size

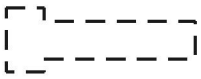


Print at 100 % (actual size). Lay a part on its dashed outline to identify it.

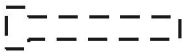
Bolts


M5×35 Screw


M3×16 Screw


M5×20 Screw


M3×8 Screw


M3×20 Screw

Nuts


M5 Nut


M3 Nut

Hex keys

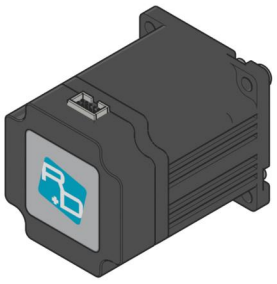

4 mm Hex key


2.5 mm Hex key

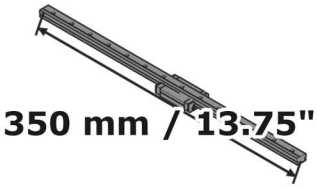

2 mm Hex key

Parts

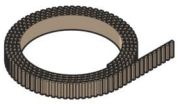
Motor and linear rail



1×
Servo motor

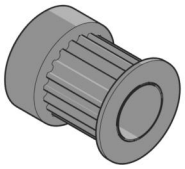


1×
Linear rail



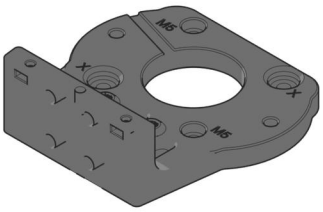
500 mm

1×
Belt

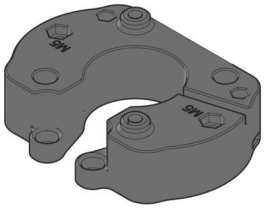


10 mm
1×
Pulley

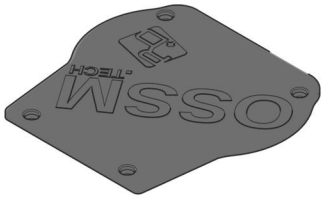
3D printed



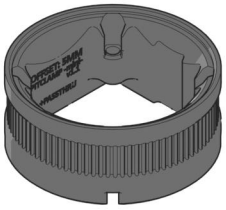
1×
Body bottom



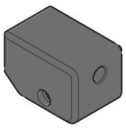
1×
Body middle



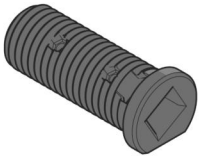
1×
Cover



1×
Motor ring



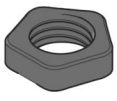
1×
Belt tensioner



1×
24 mm
Threaded end effector



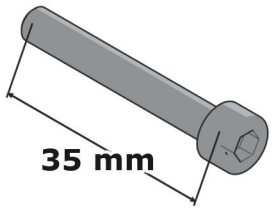
1×
Belt clamp



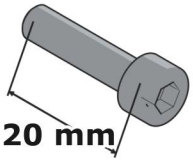
1×
Lock nut

Parts

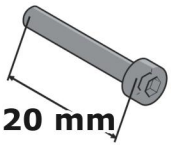
Bolts



4×
M5×35 Screw



3×
M5×20 Screw



1×
M3×20 Screw



2×
M3×16 Screw



8×
M3×8 Screw

Nuts



5×
M5 Nut



7×
M3 Nut

Bearings



6×
5×11×4
Bearing

Tools

Required



4 mm Hex key

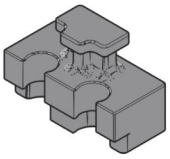


2.5 mm
Hex key



2 mm Hex key

3D printed jigs

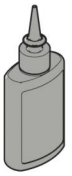


Bearing jig



3.8 mm
Pulley spacer

Recommended

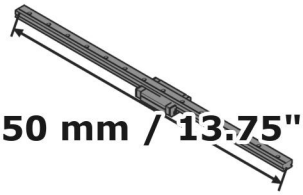


Threadlocker
(blue 243)

Legend

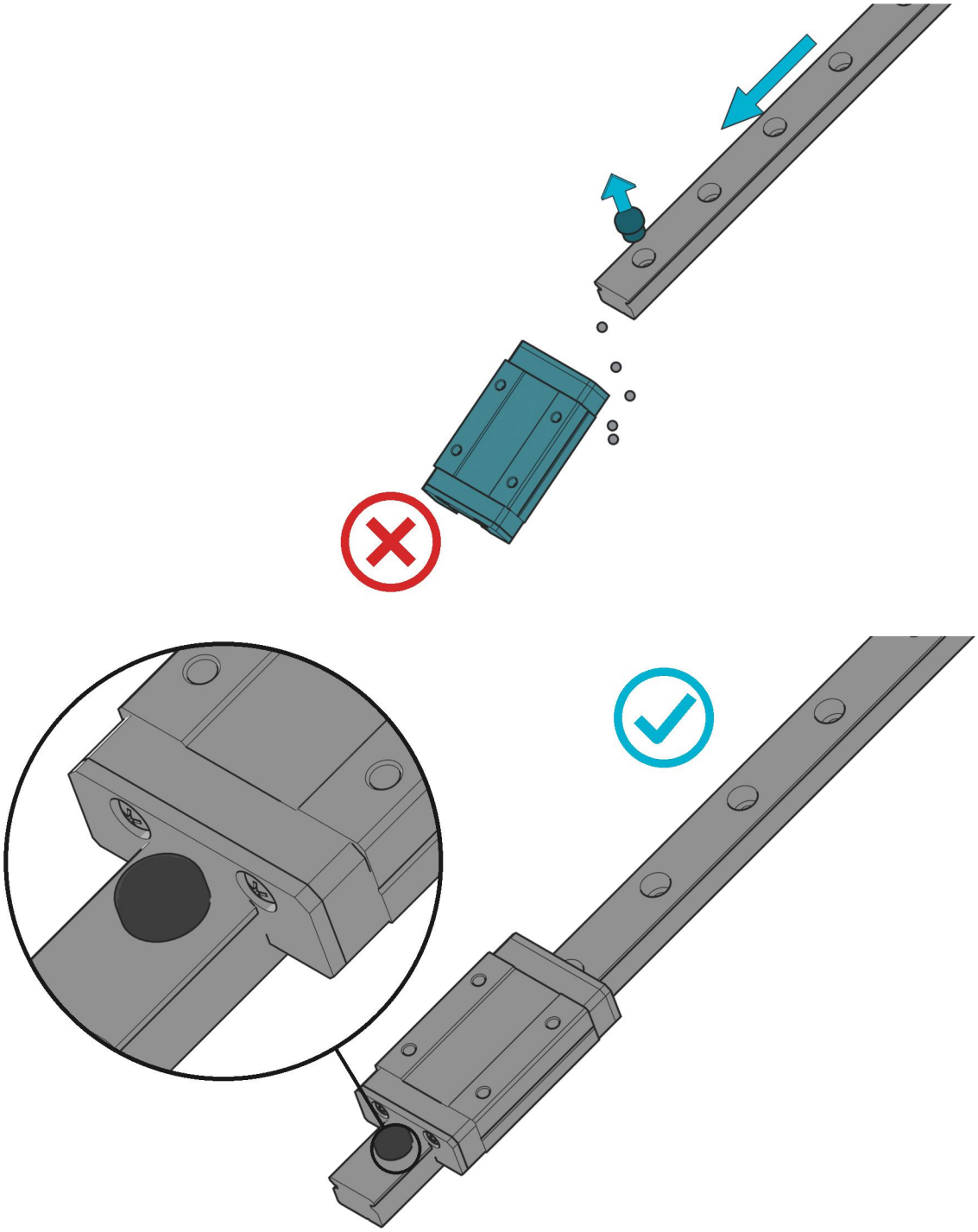
In a step's parts box, this drop beside a screw's quantity means:
put threadlocker (blue 243) on the screw's thread before it goes in.

Warnings ⚠



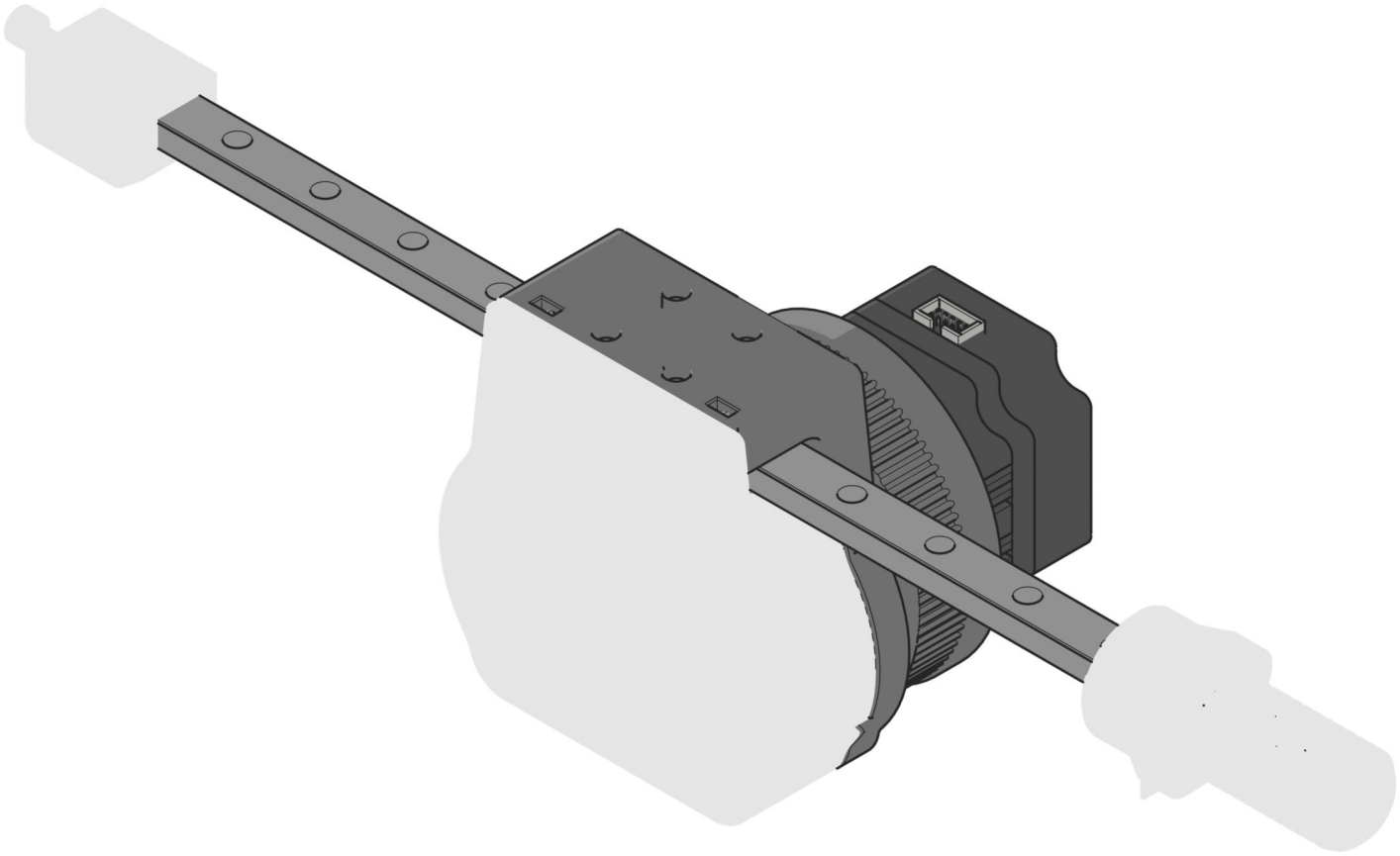
1×

Linear rail

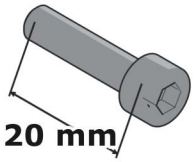


⚠ **Never let the carriage slide off the rail: its balls fall out.**

1 The motor head

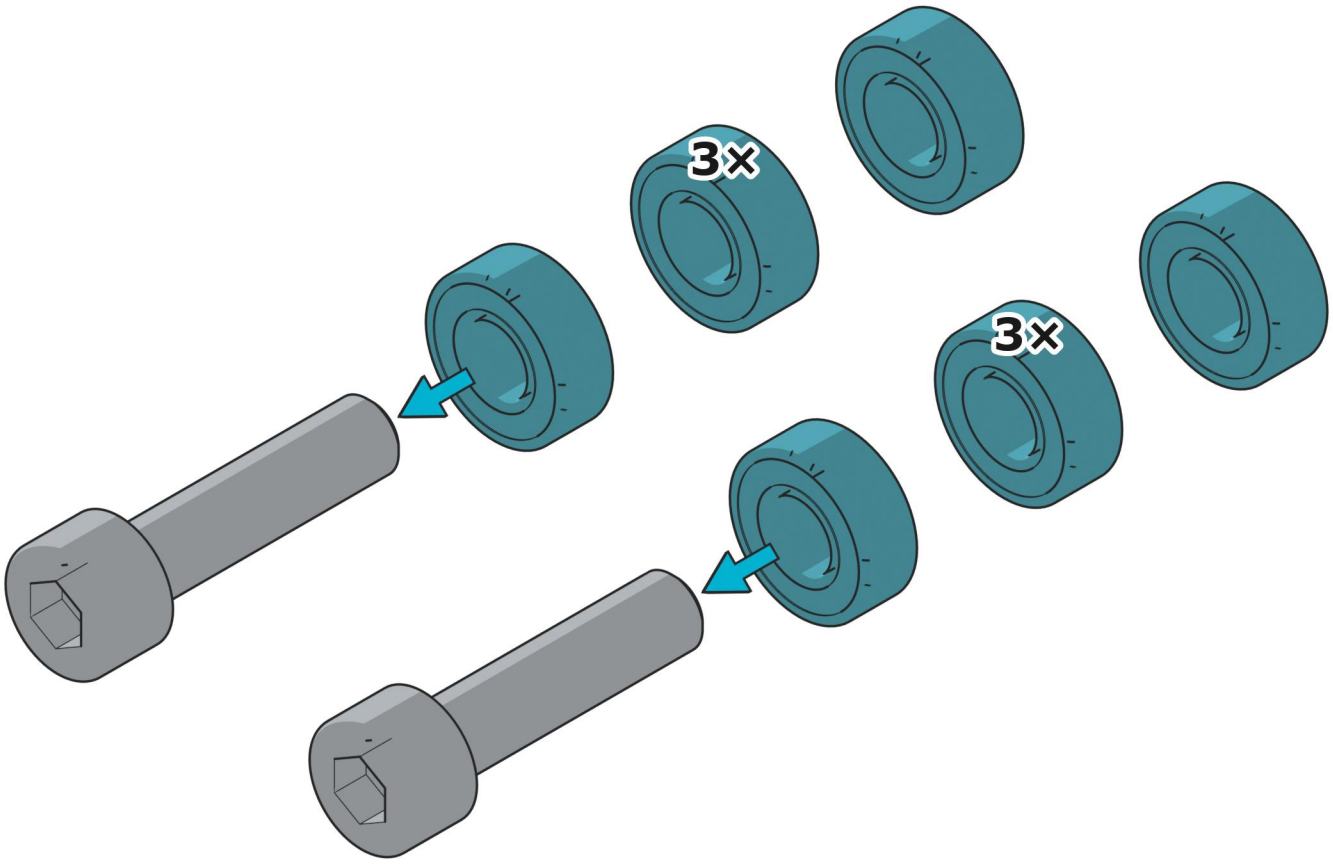


1

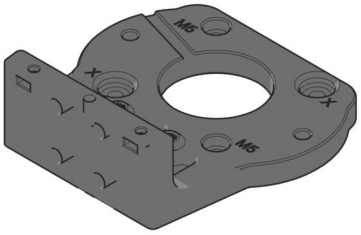


2×
M5×20 Screw

6×
5×11×4 Bearing



2

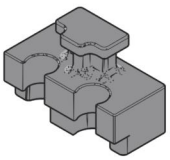


1×

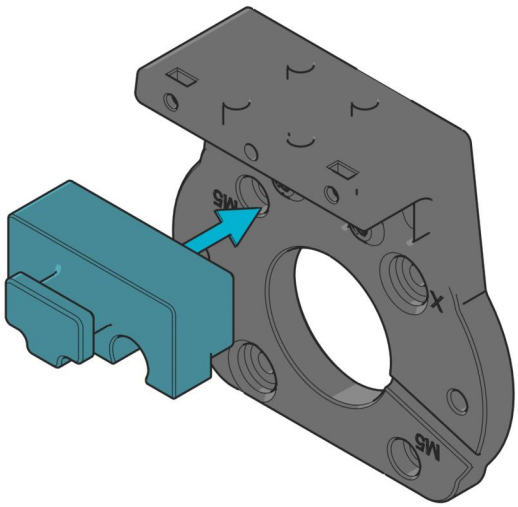
Body bottom

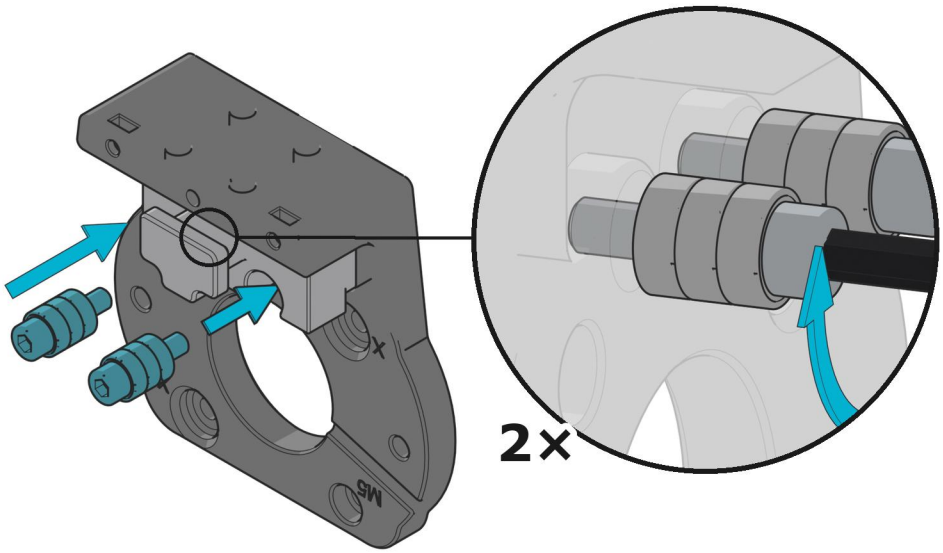


4 mm Hex key



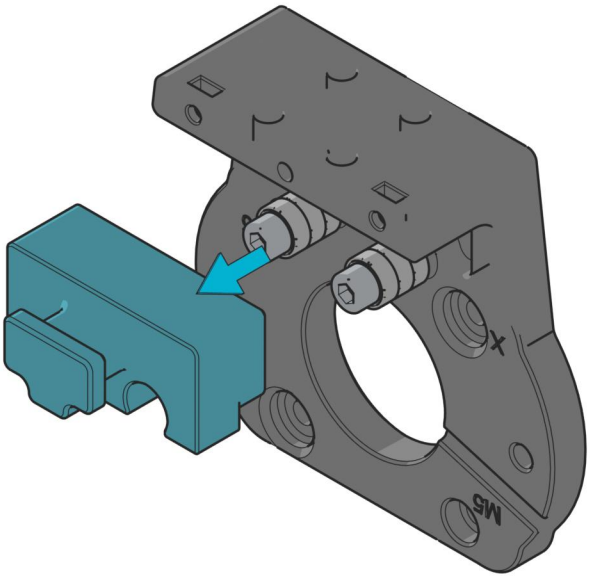
Bearing jig

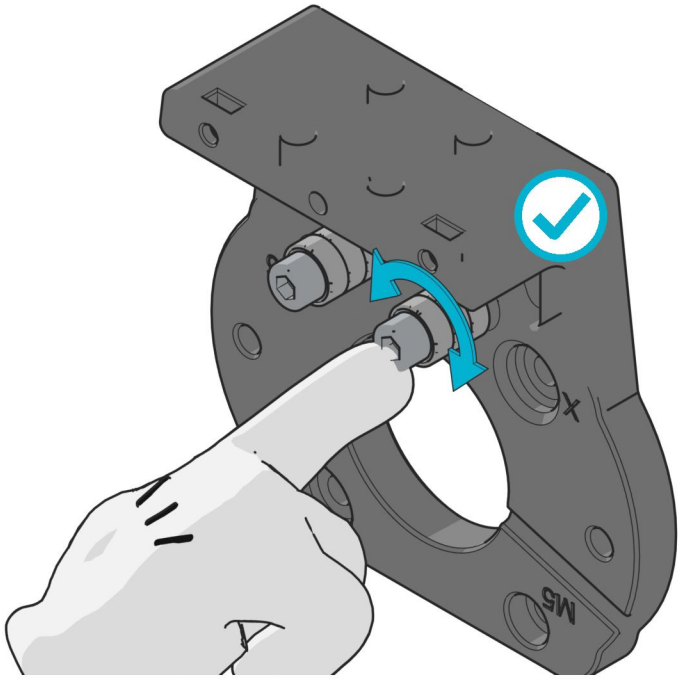




Snug only: do not over-tighten the idler screws.

3





Snug only: each bearing must spin freely. If one does not, loosen its screw a little.

4

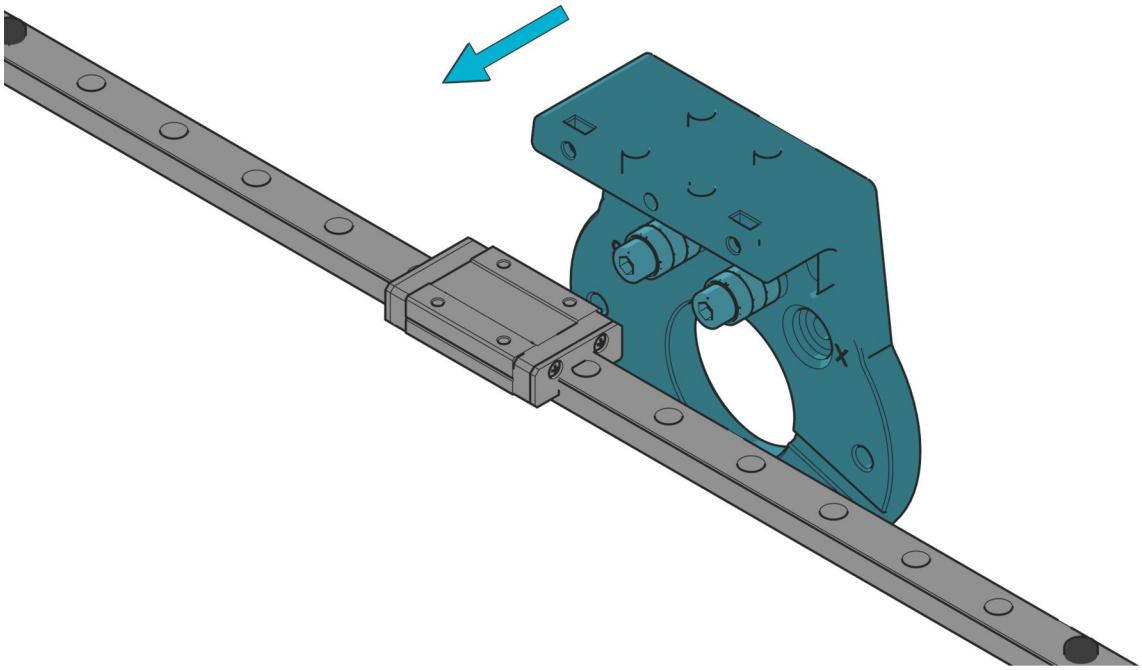


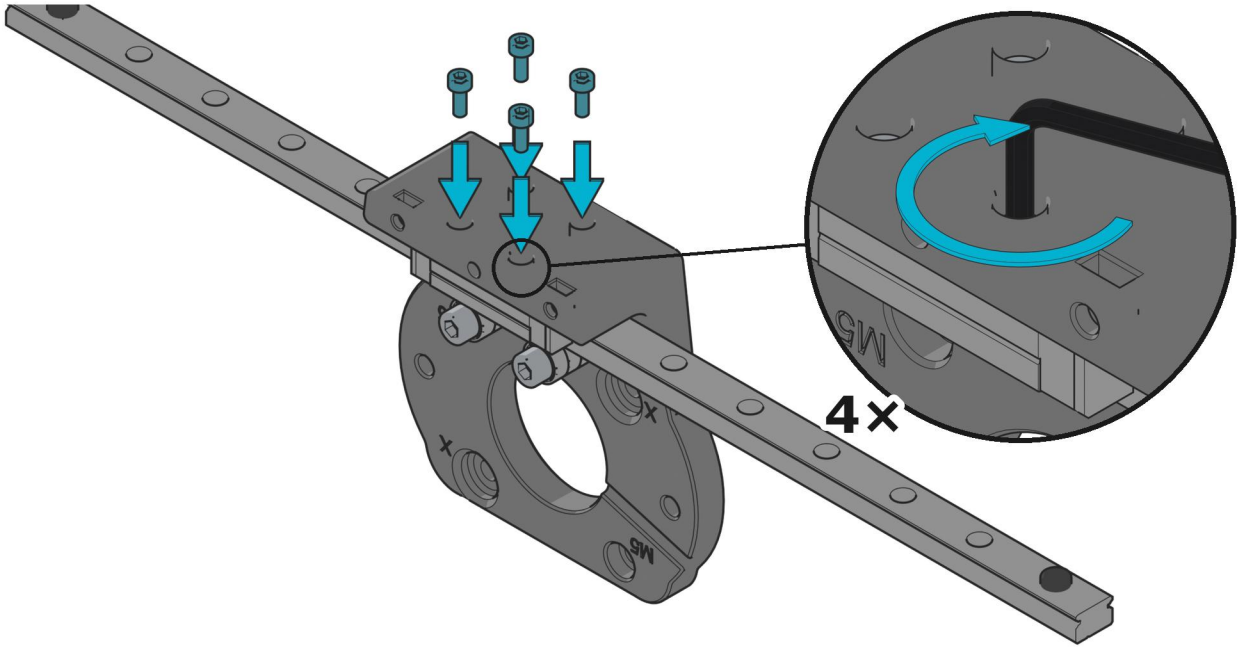
4× M3×8 Screw

1× Linear rail

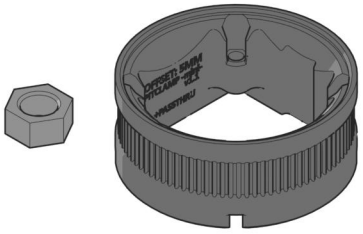


2.5 mm Hex key



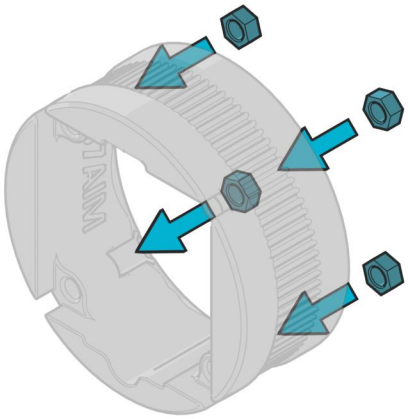


5

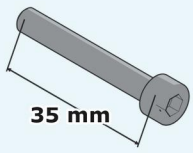


4× M5 Nut

1× Motor ring



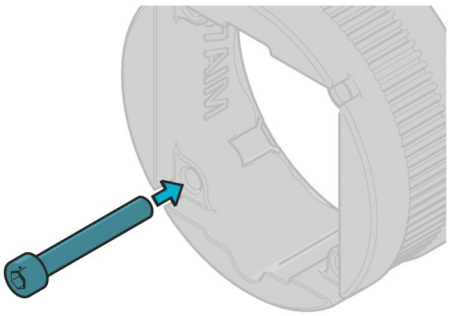
Tip: pulling the nuts into place

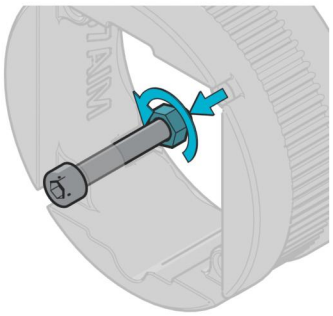


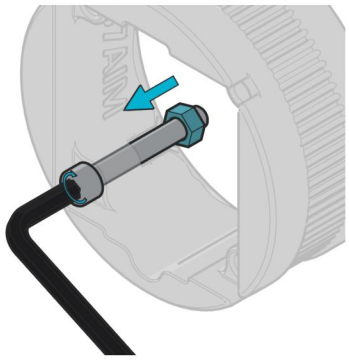
1× M5×35 Screw

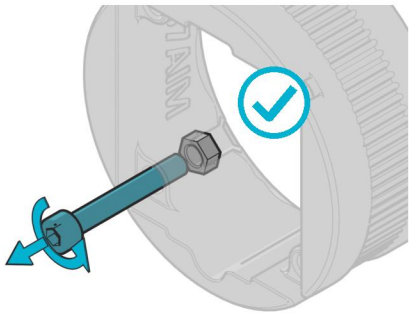


4 mm Hex key







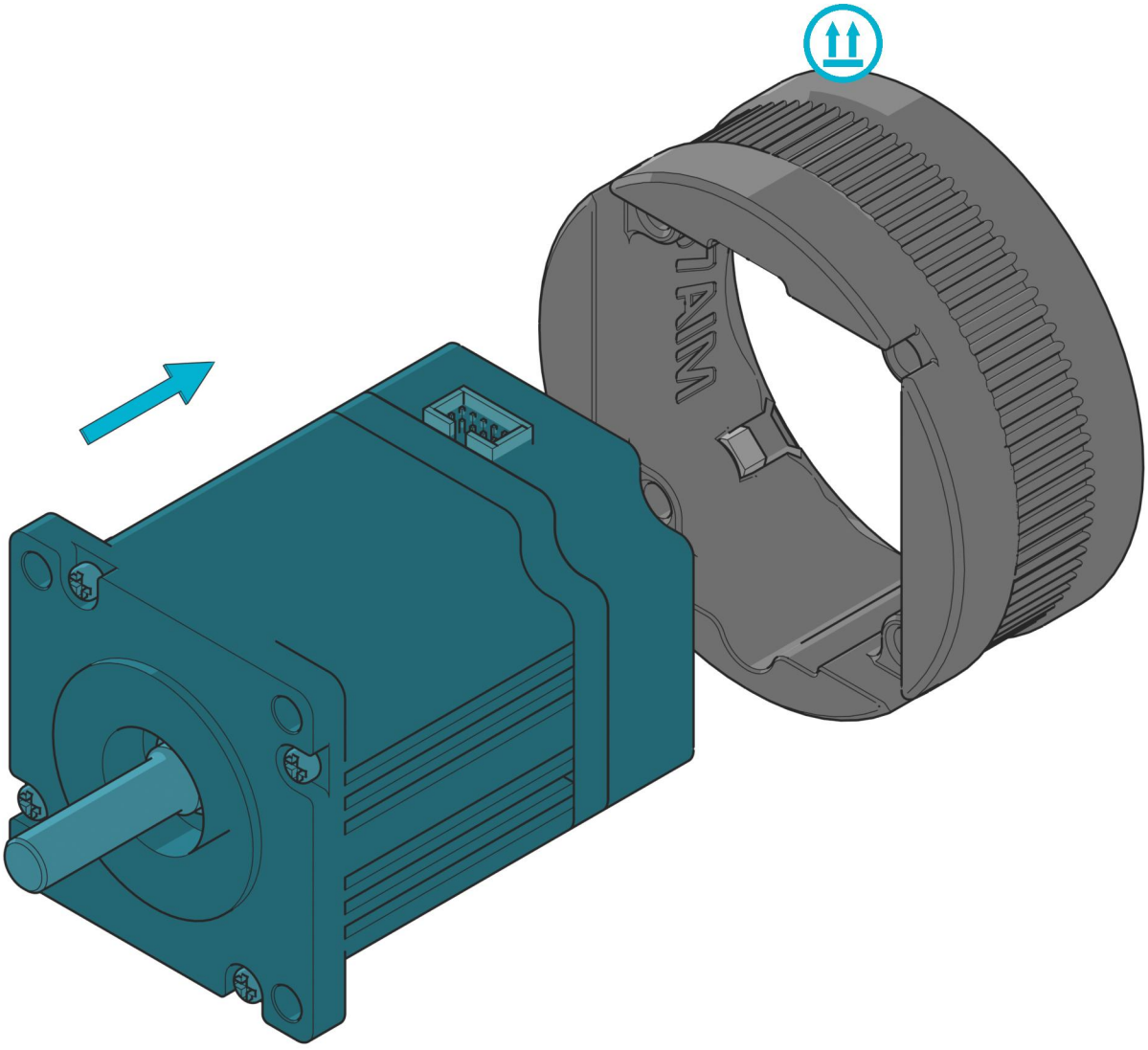


6

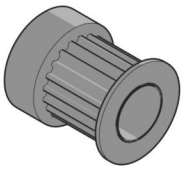


1×

Servo motor



7



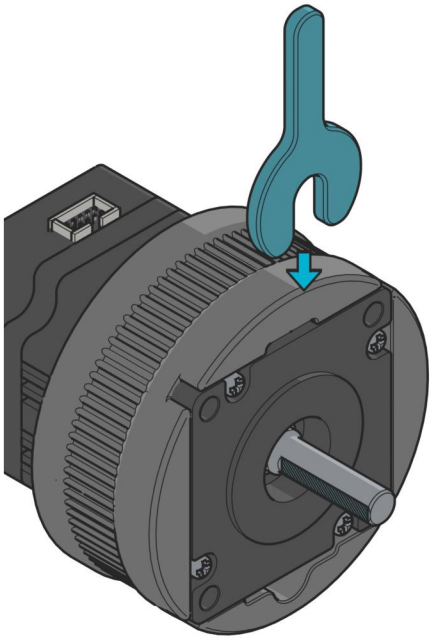
10 mm

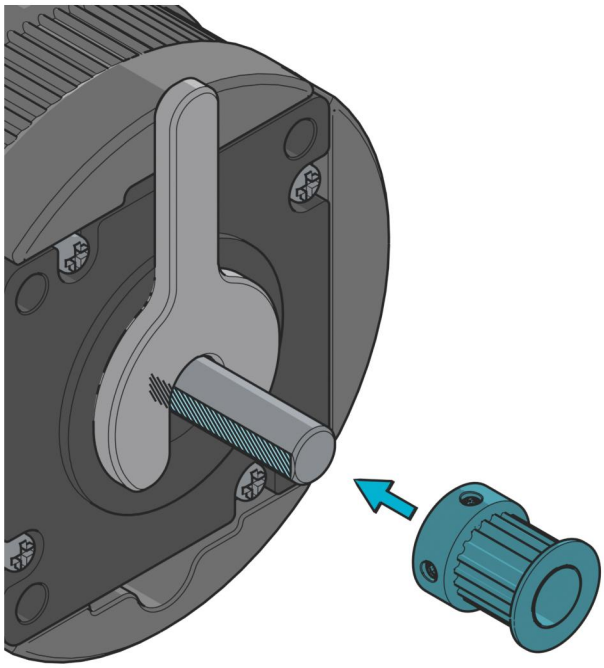
1×

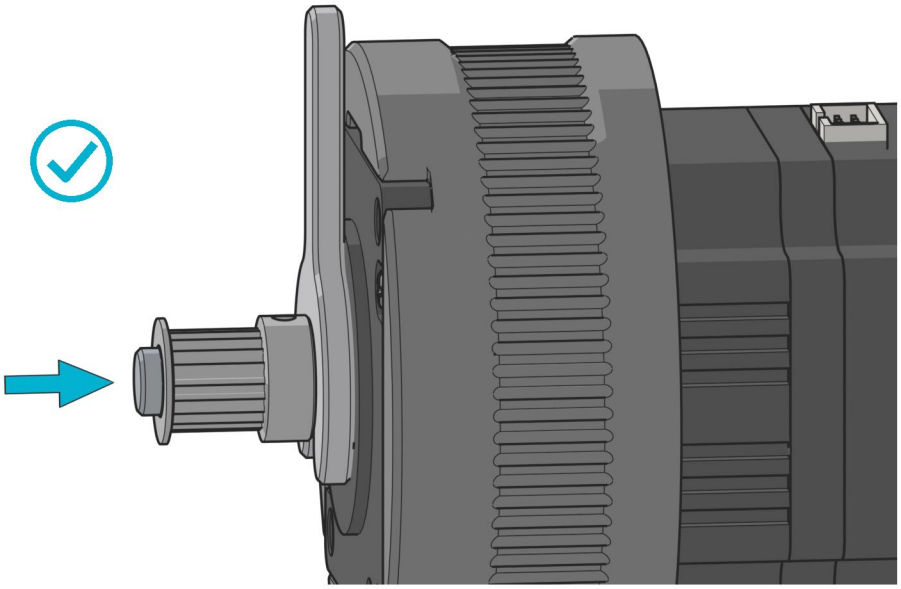
Pulley



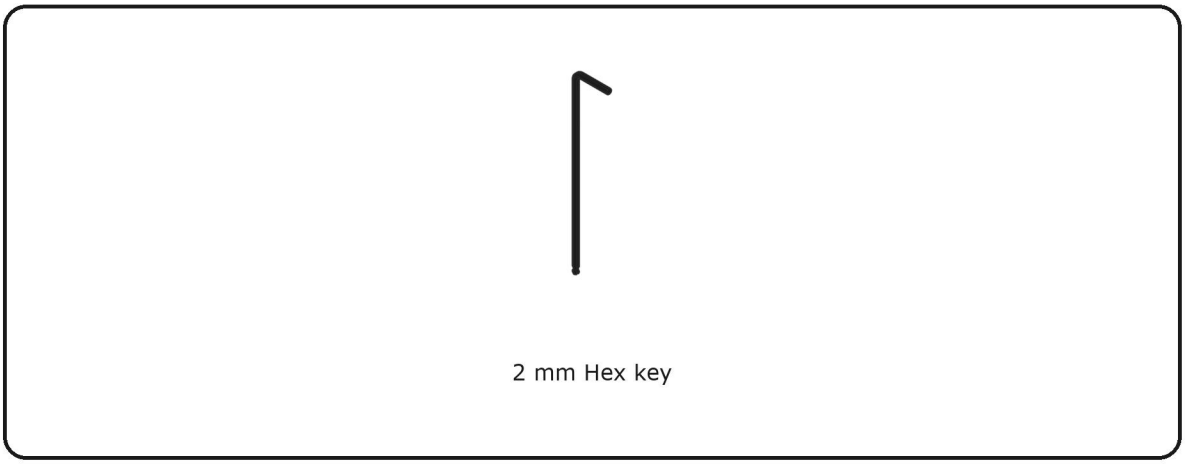
3.8 mm
Pulley spacer

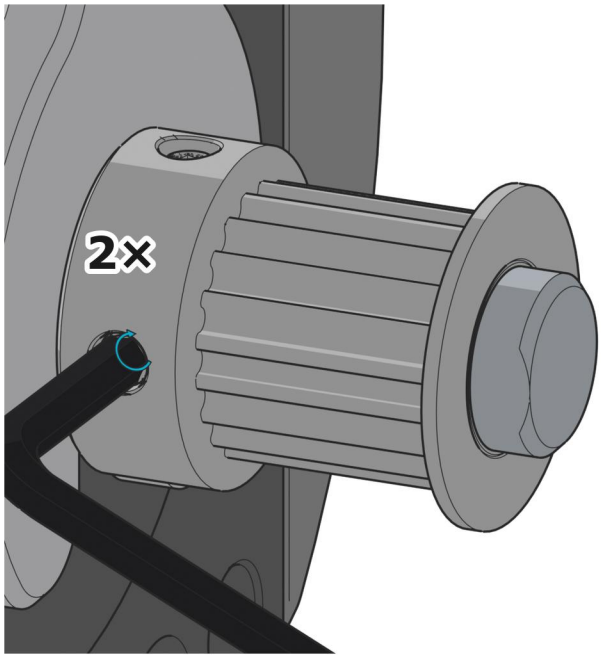


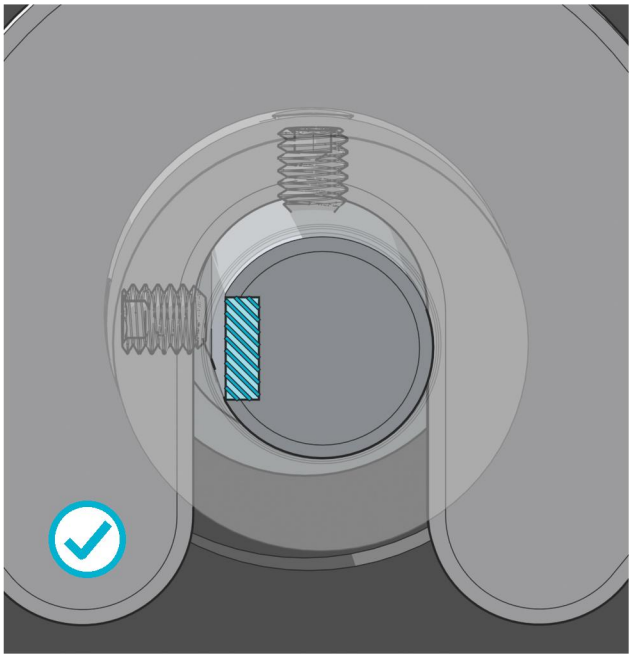


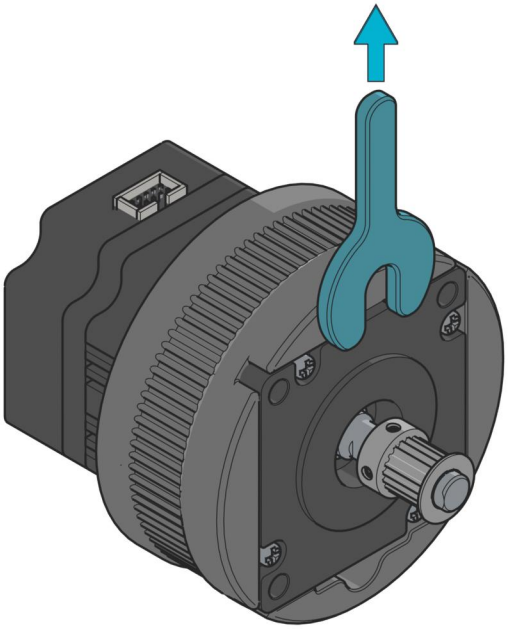


8



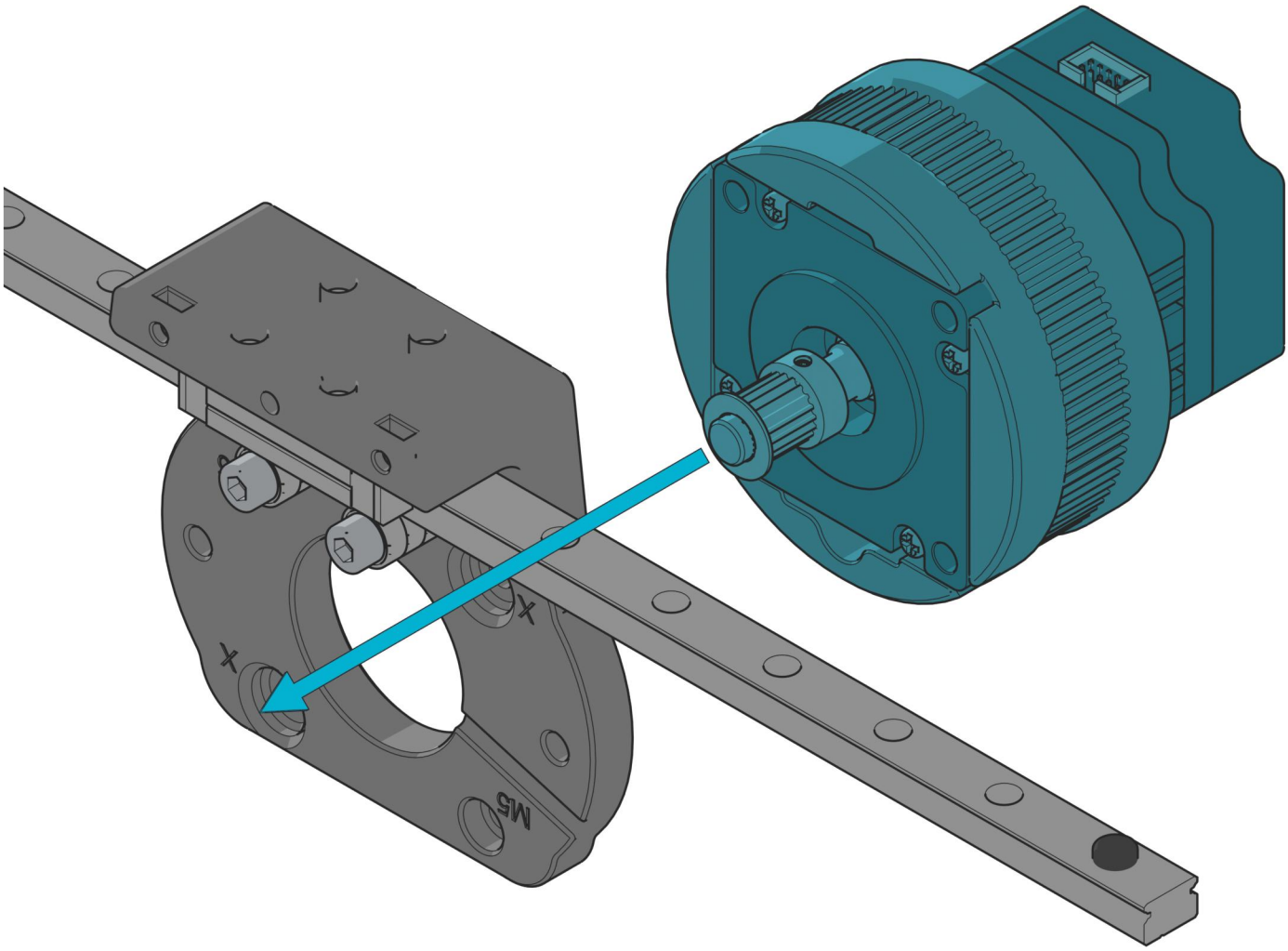




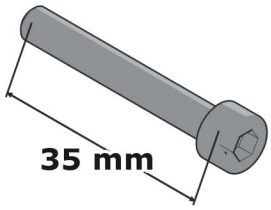


Keep the pulley pushed against the spacer while you tighten.

9



10

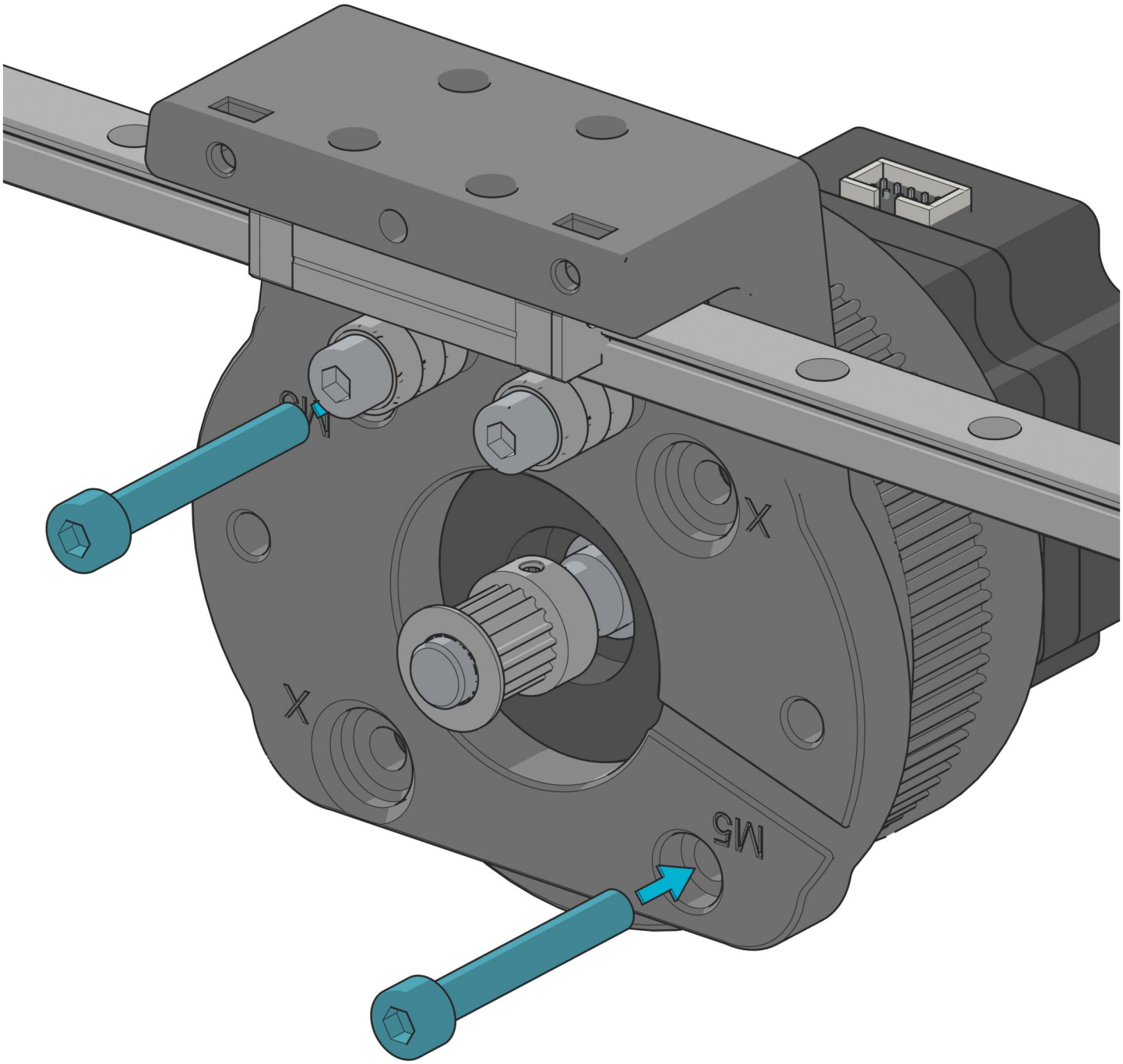


2×

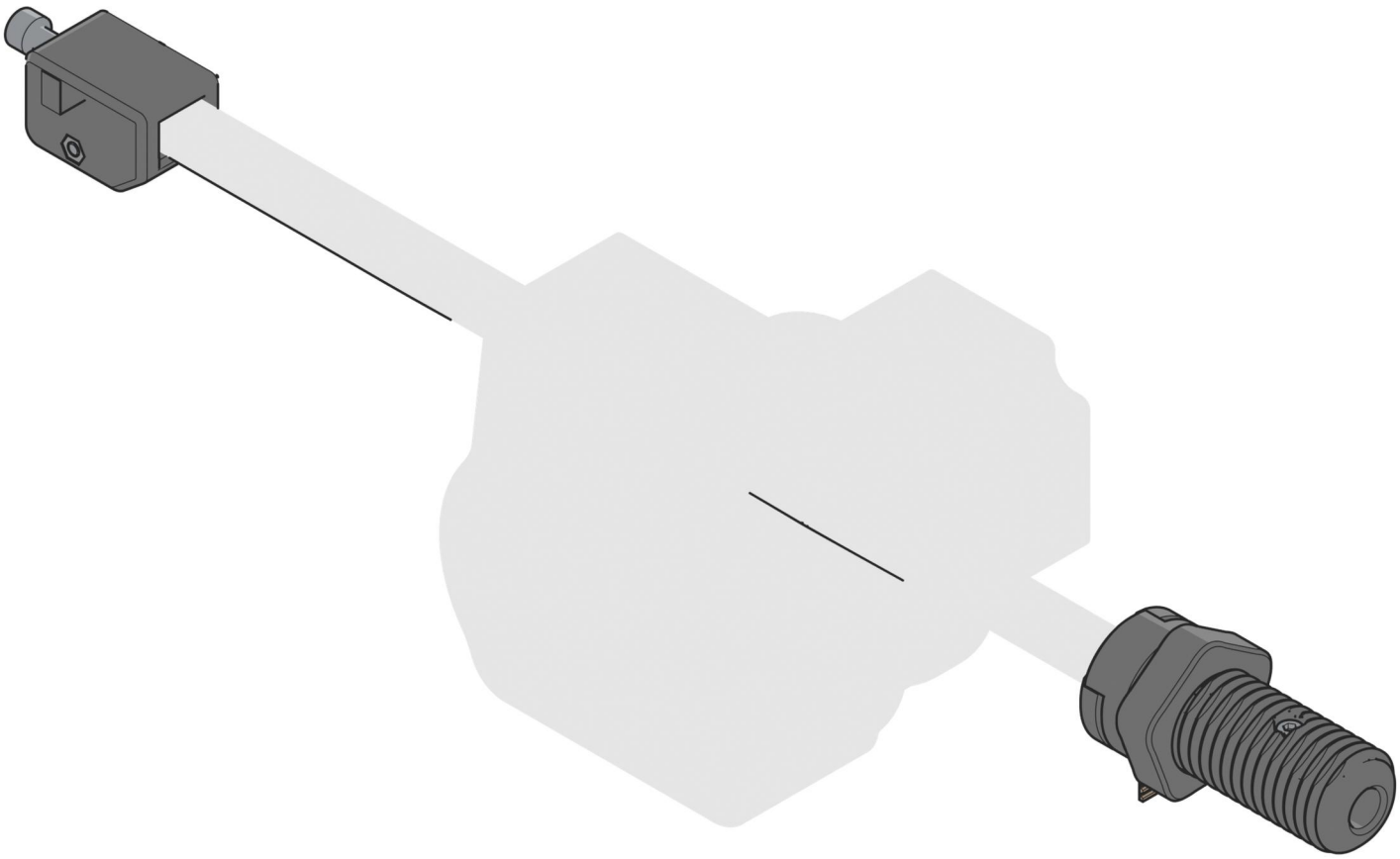
M5×35 Screw



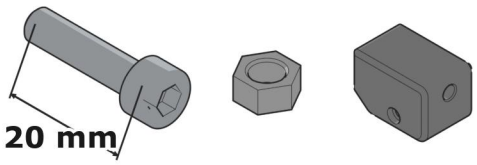
4 mm Hex key



2 The belt



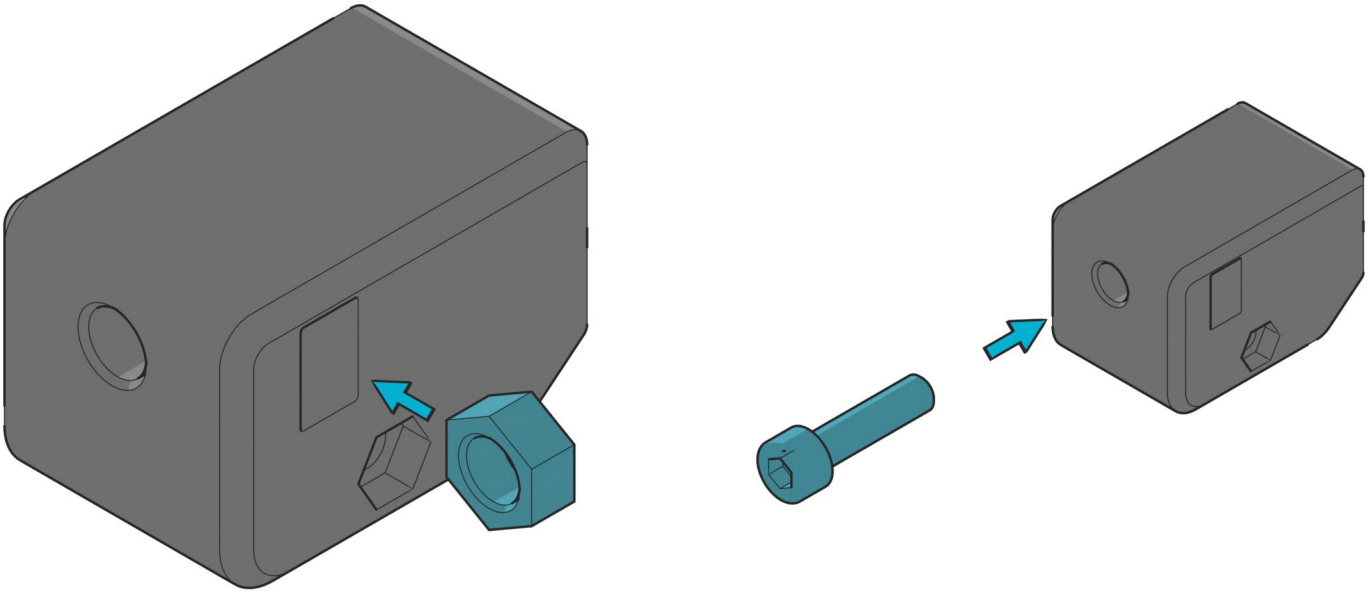
11

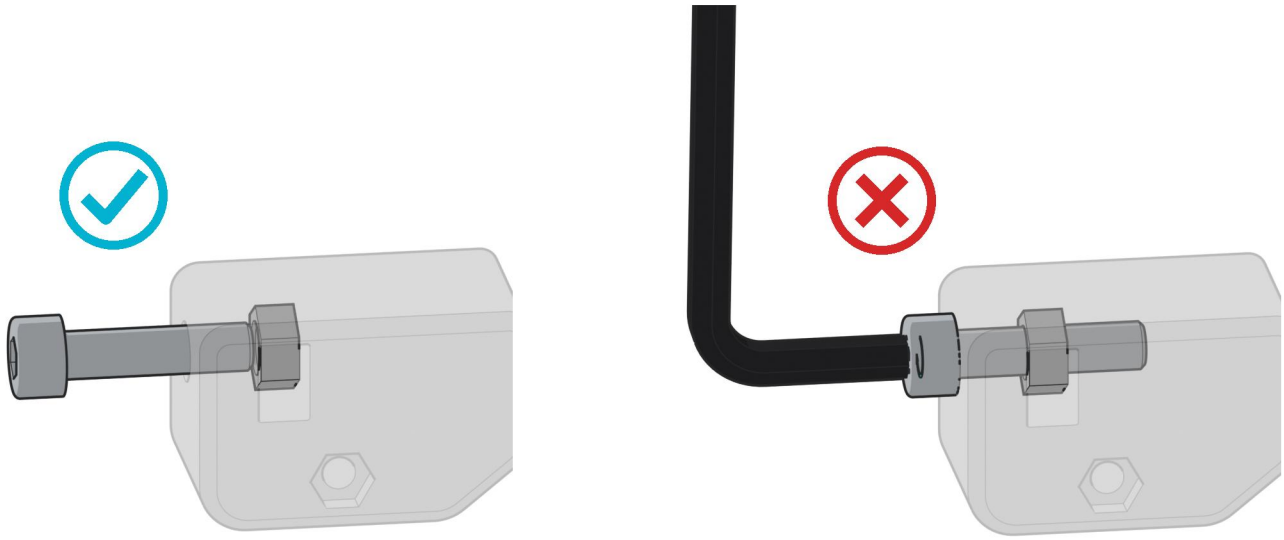


1× M5×20 Screw

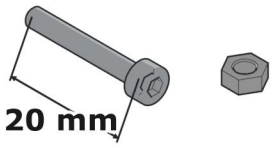
1× M5 Nut

1× Belt tensioner





12

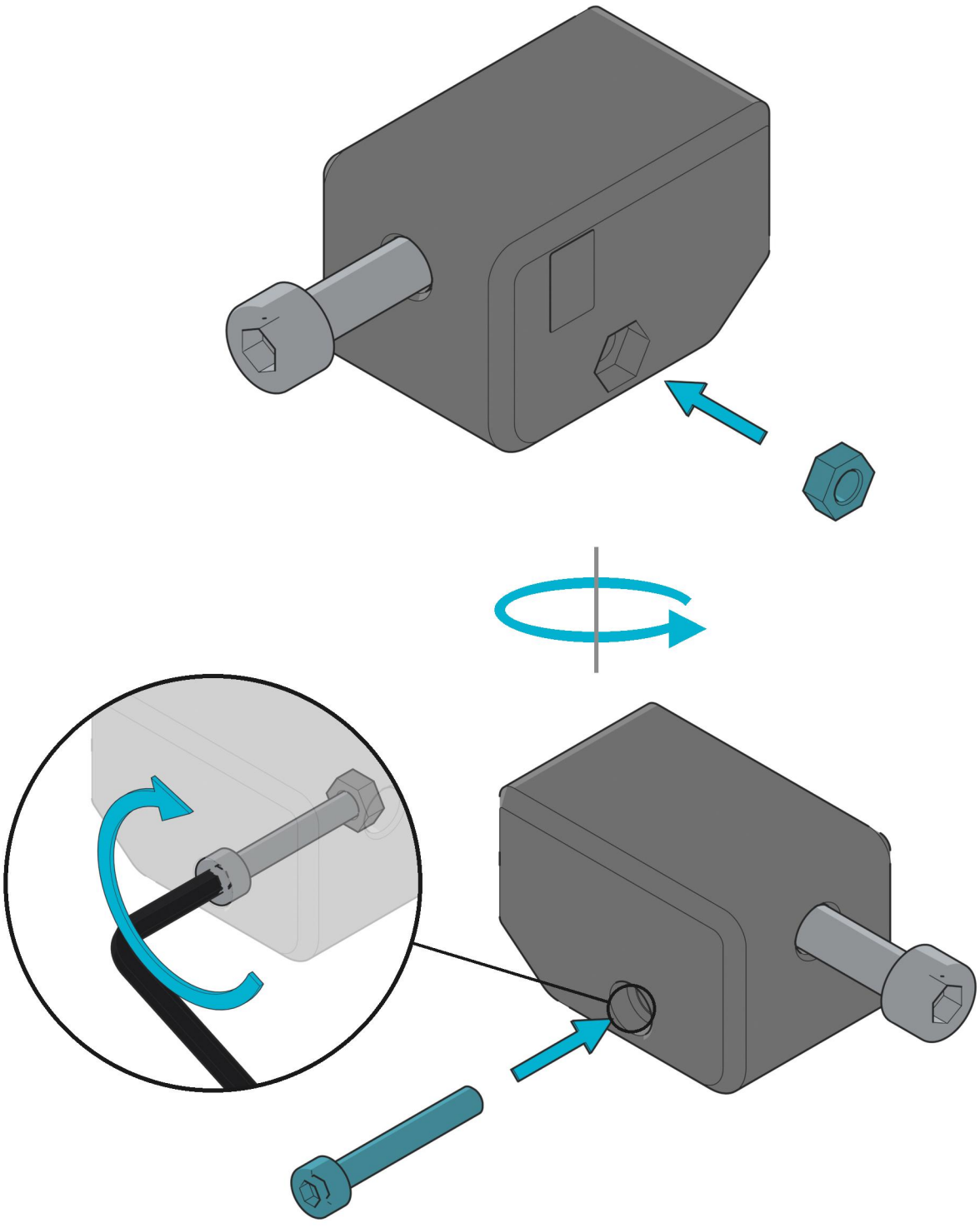


1× M3×20 Screw

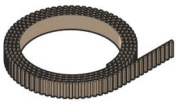
1× M3 Nut



2.5 mm Hex key



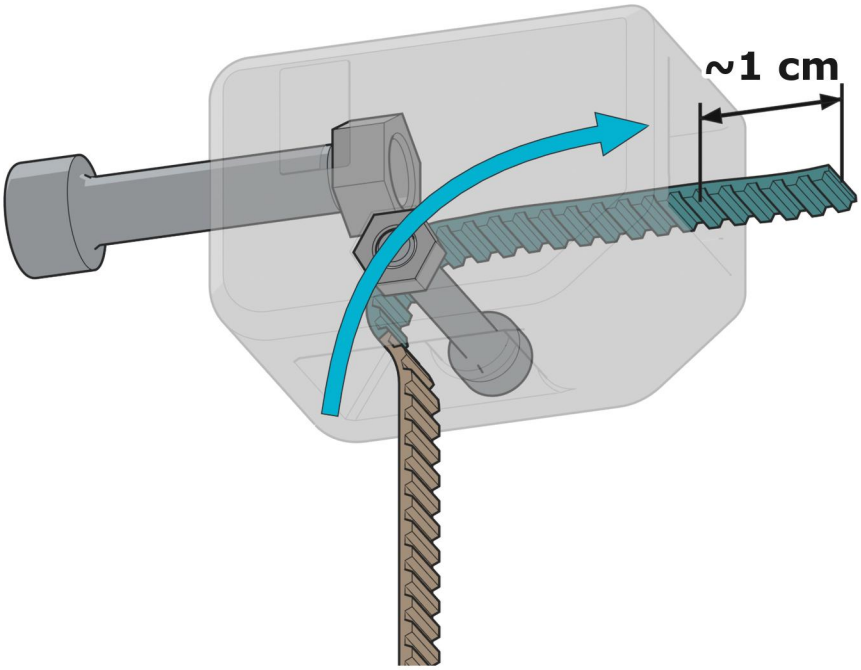
13

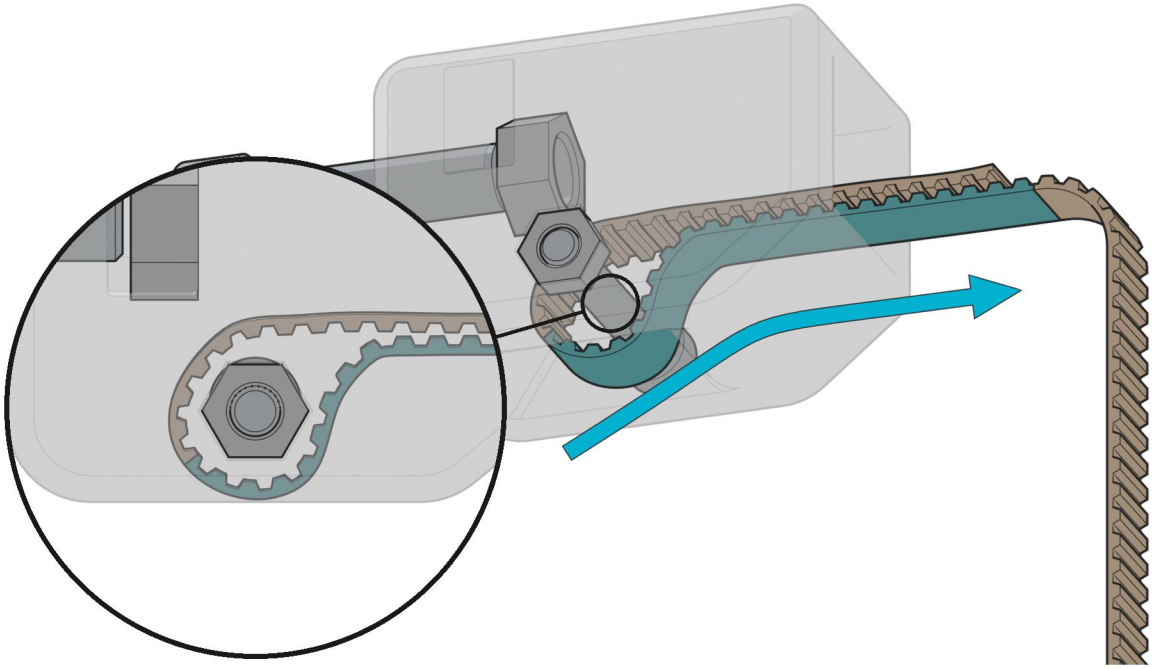


500 mm

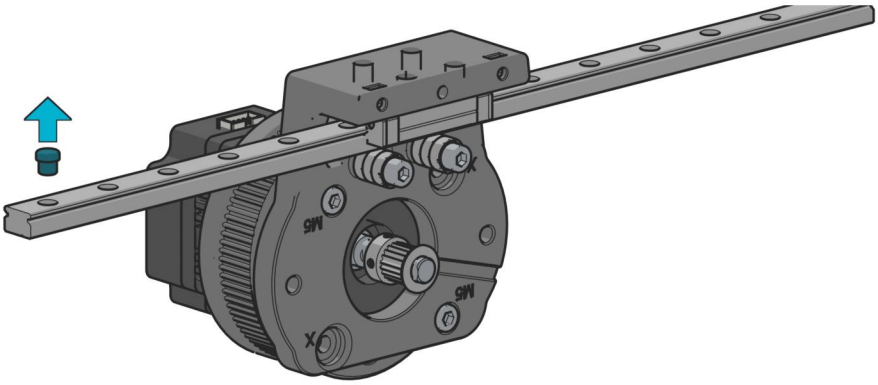
1×

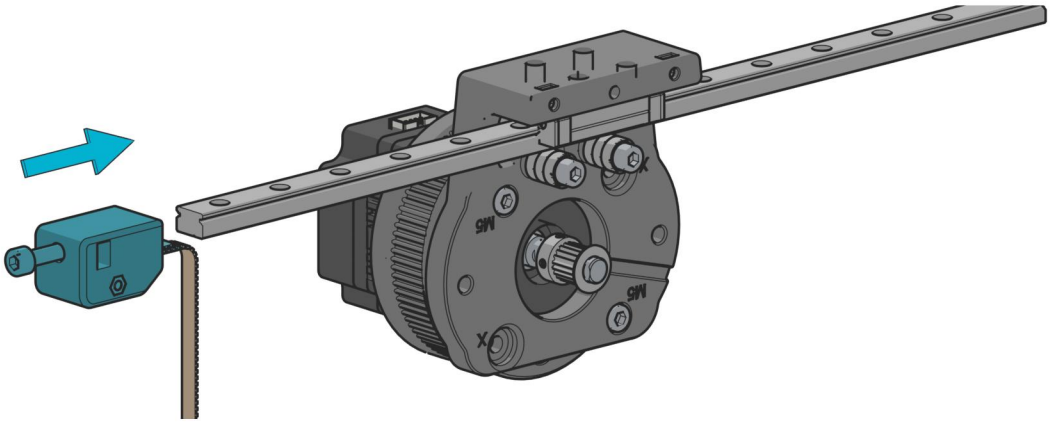
Belt

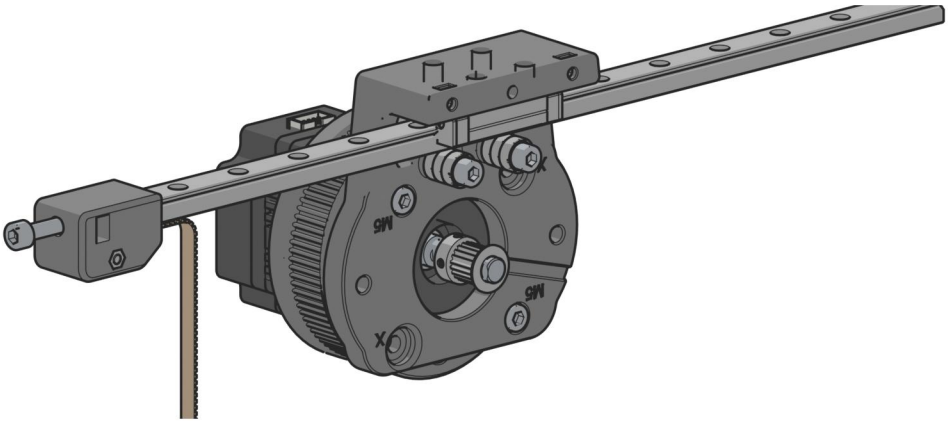




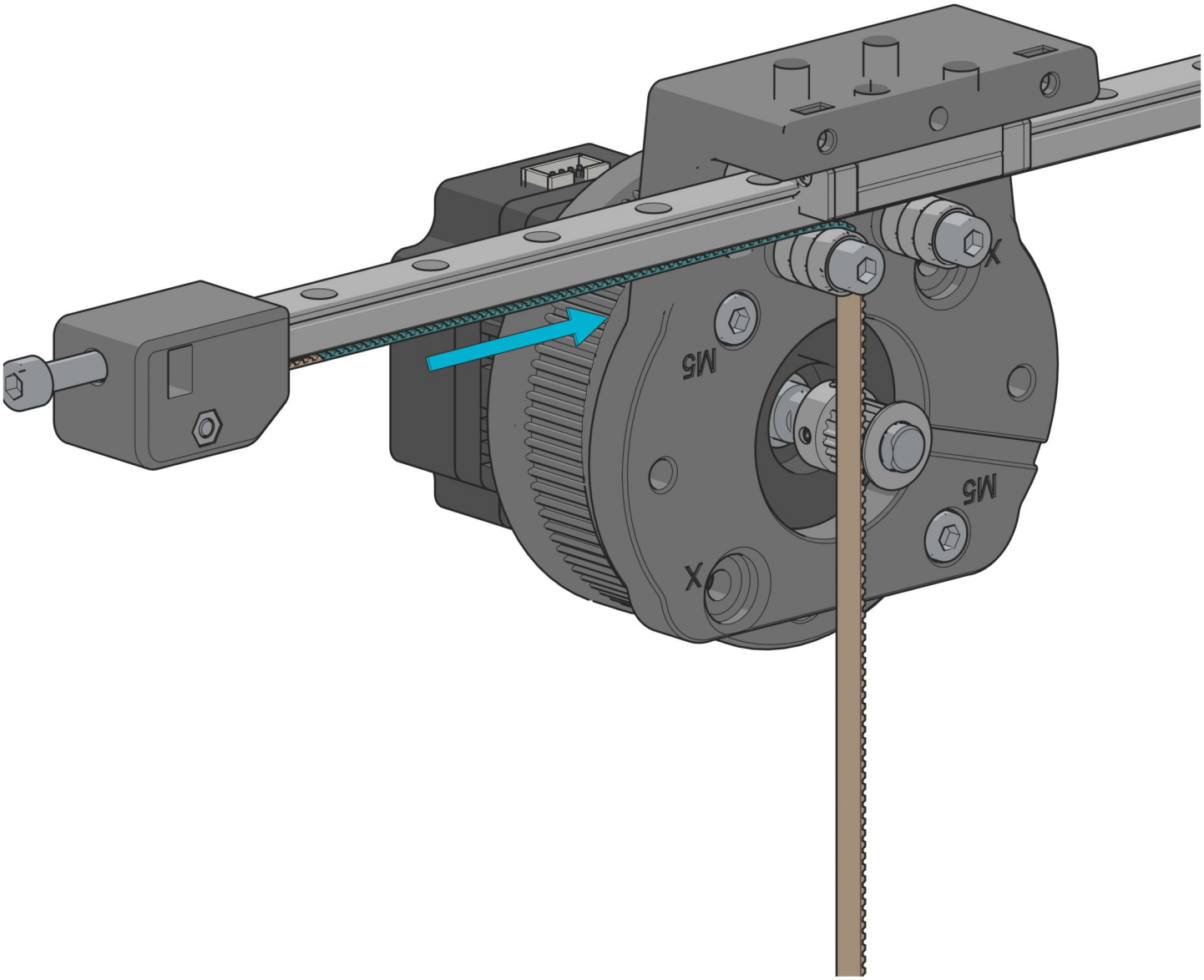
14



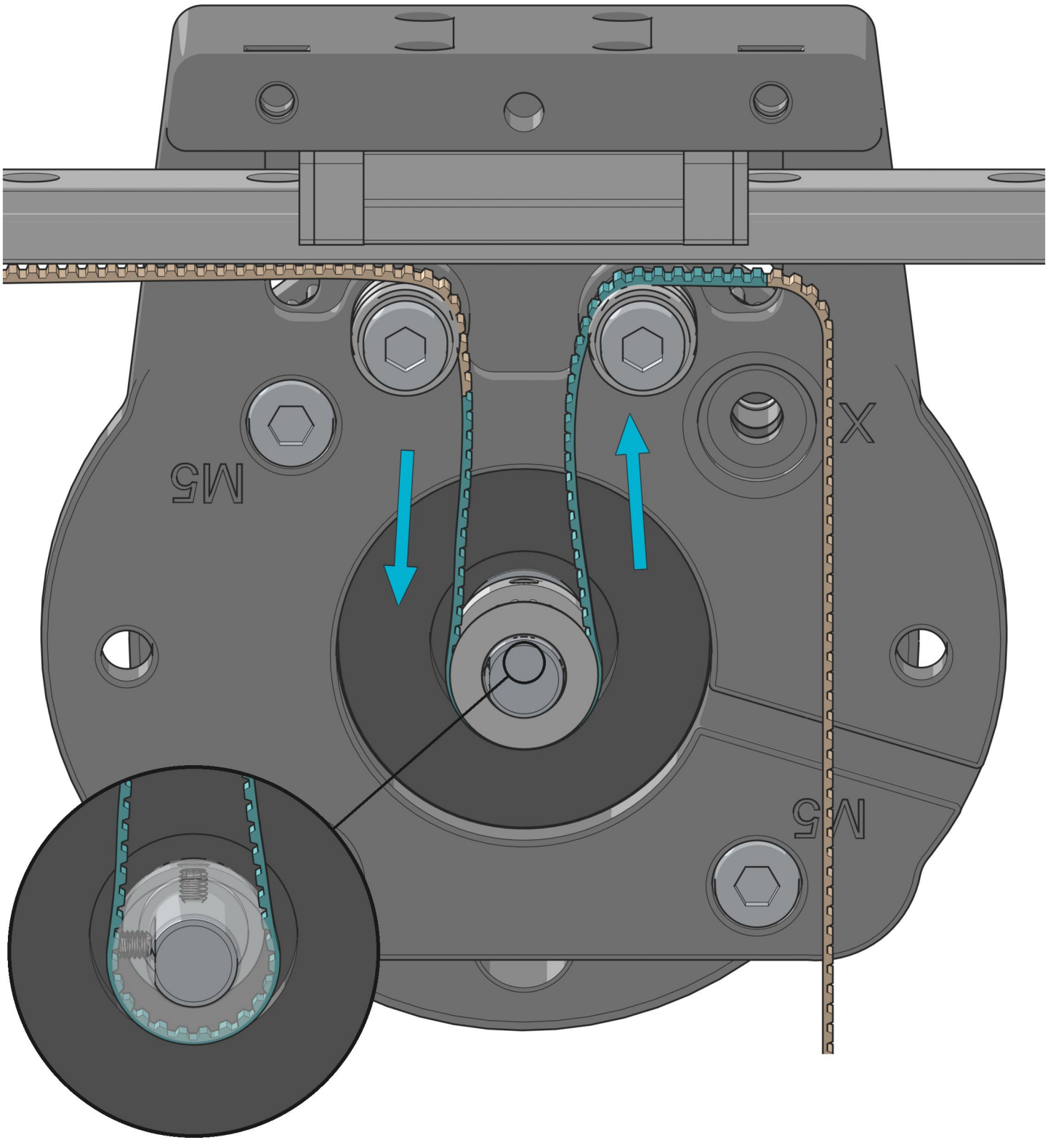




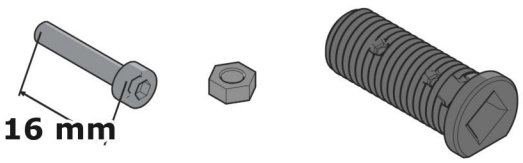
15



16



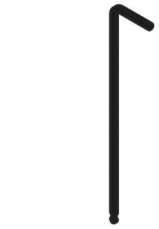
17



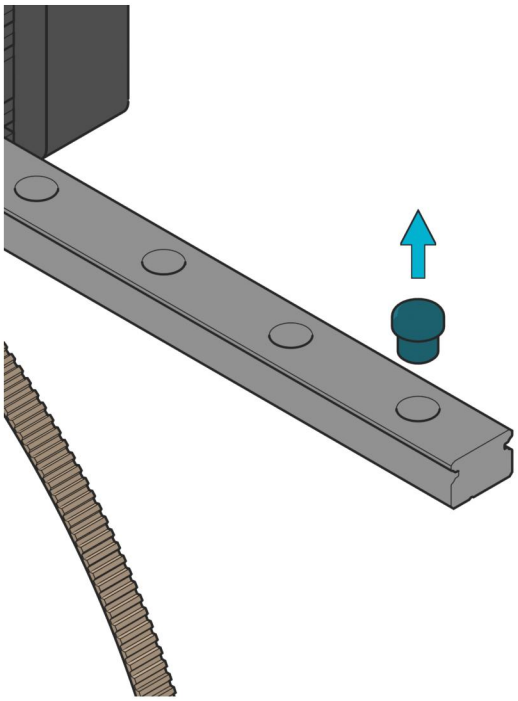
2× M3×16 Screw

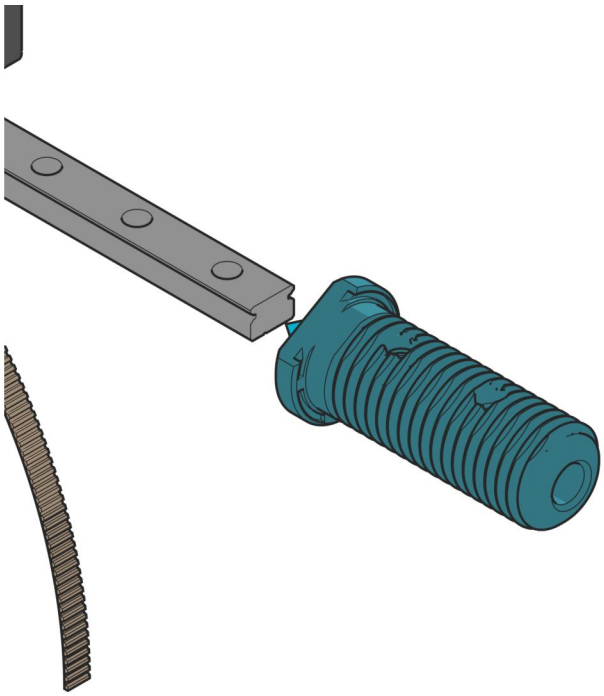
2× M3 Nut

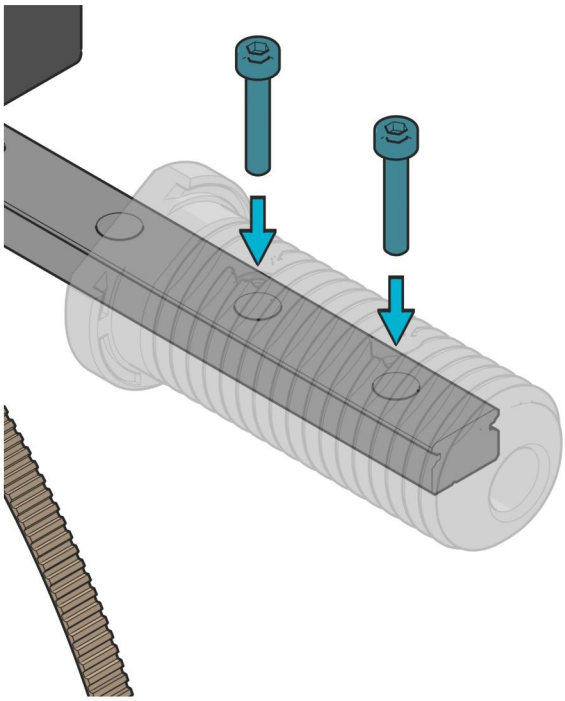
1× 24 mm Threaded end effector

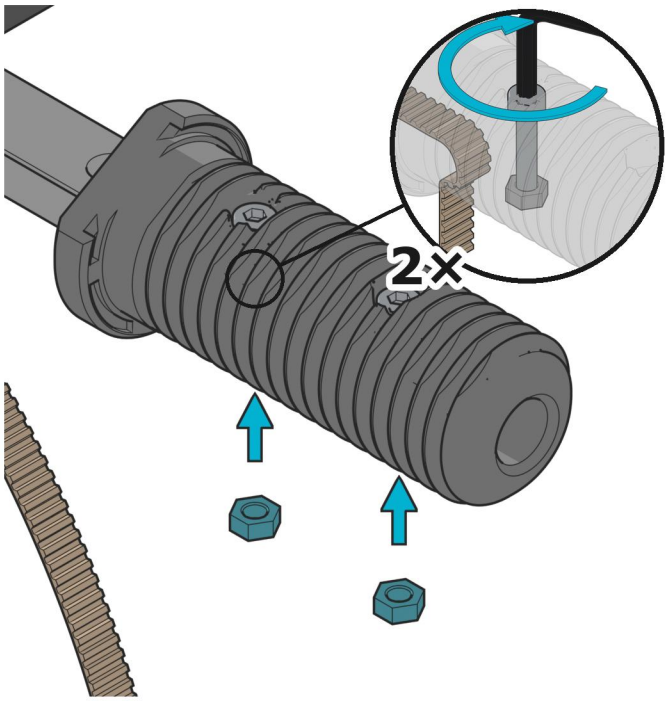


2.5 mm Hex key

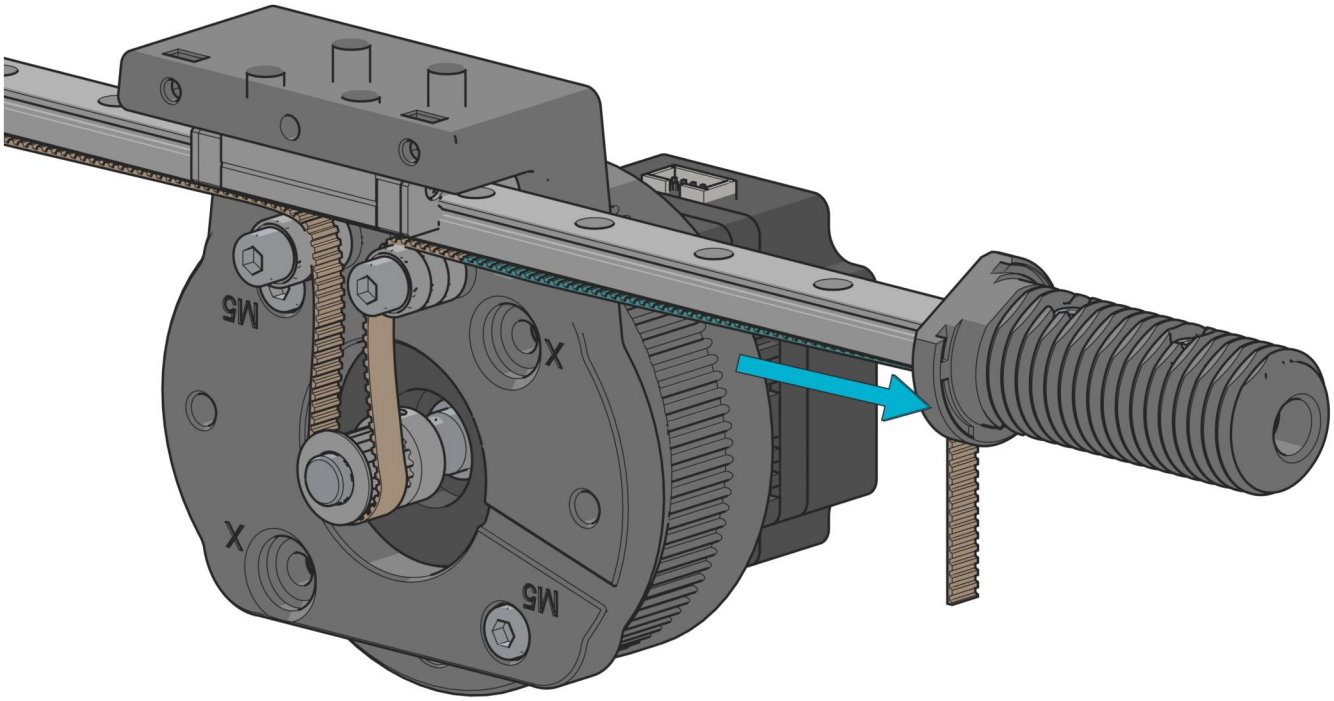




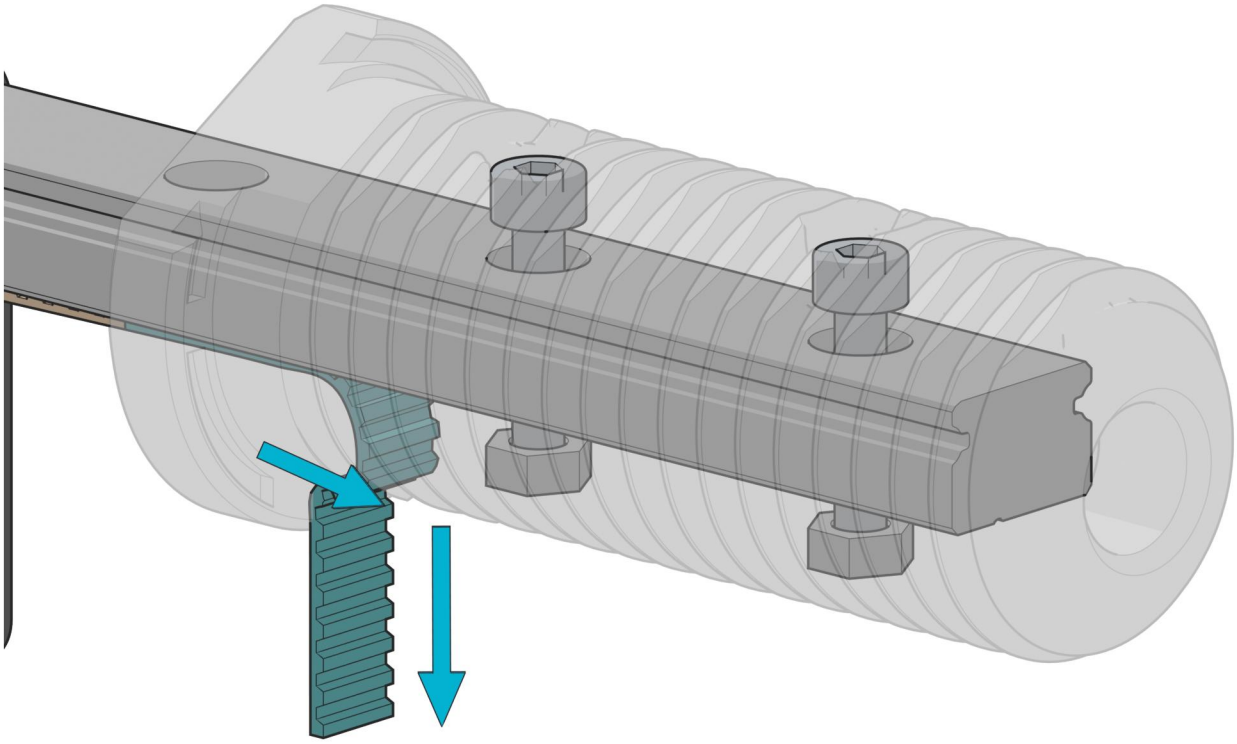




18



19

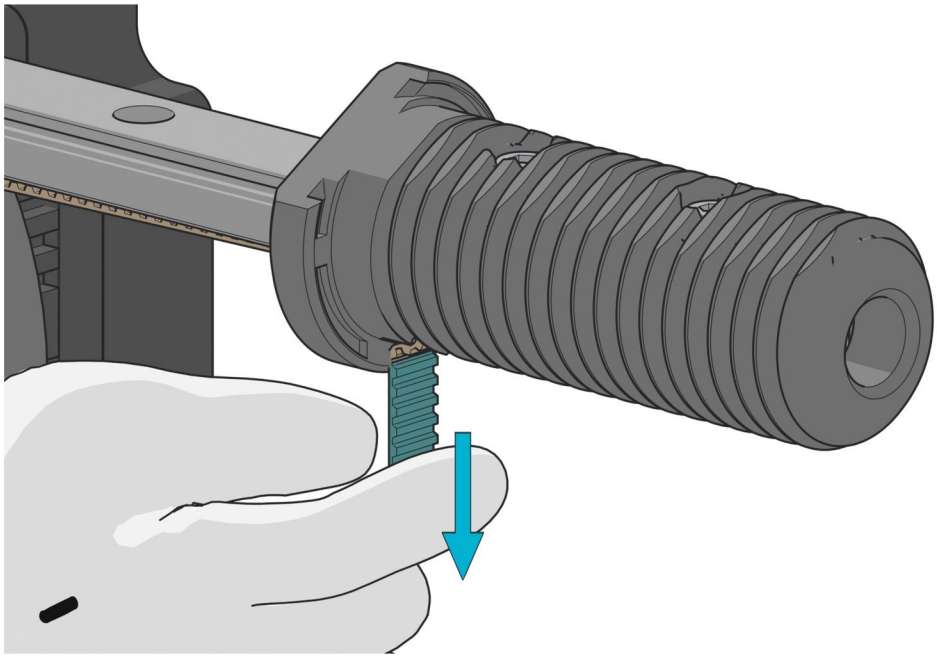


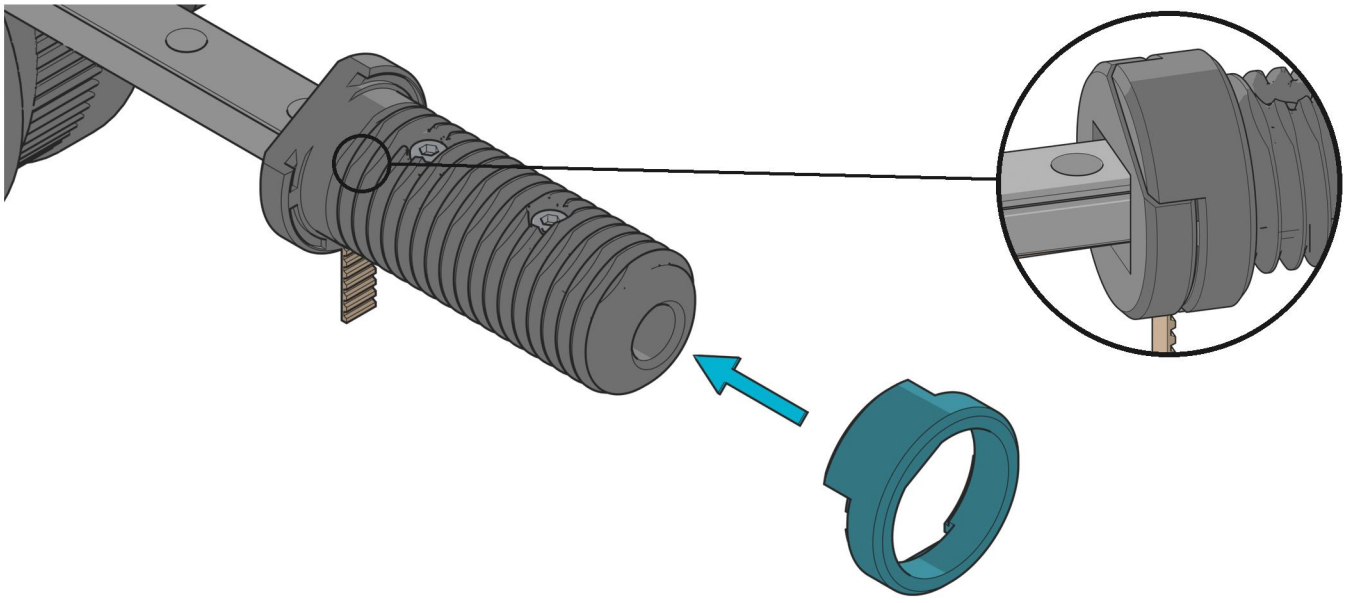
20



1×

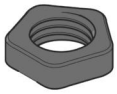
Belt clamp





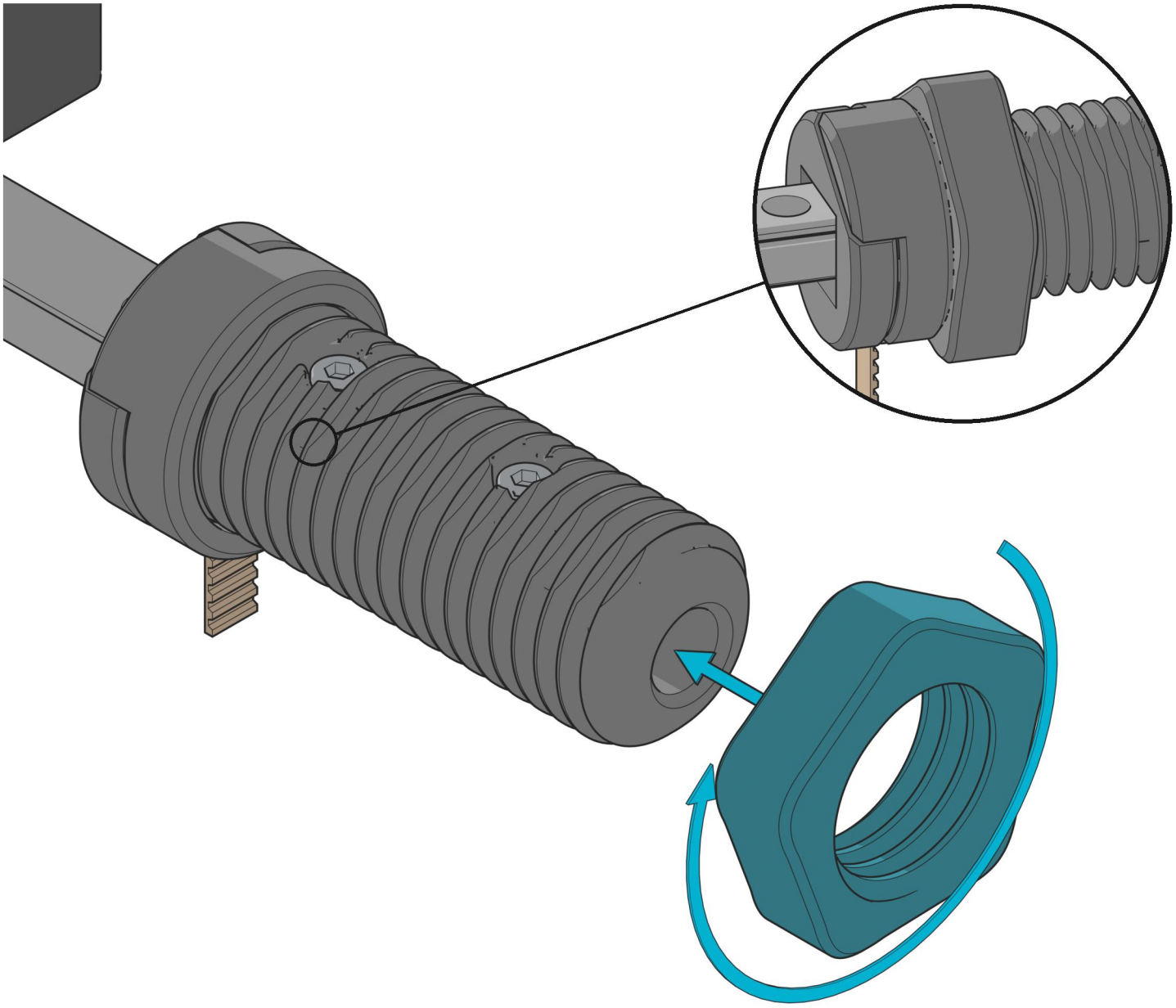
Pull the belt's end down until the belt is taut.

21

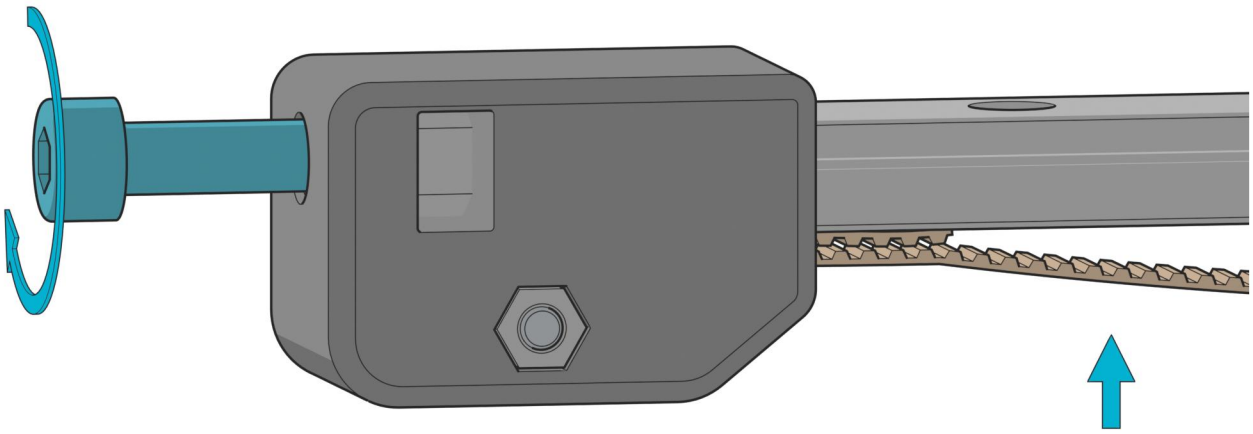


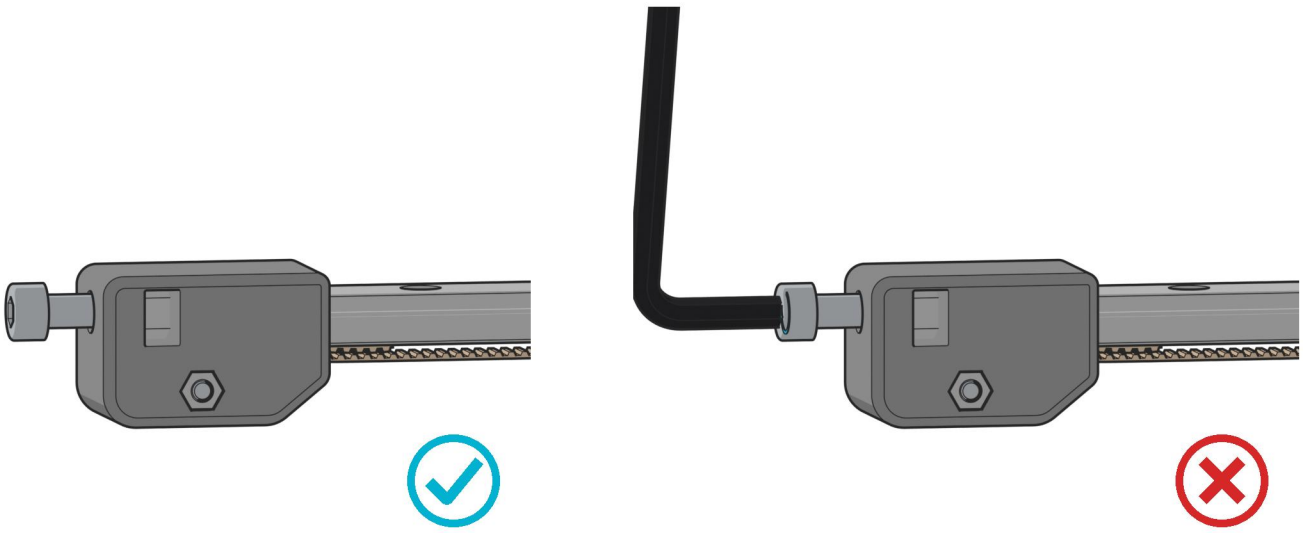
1×

Lock nut



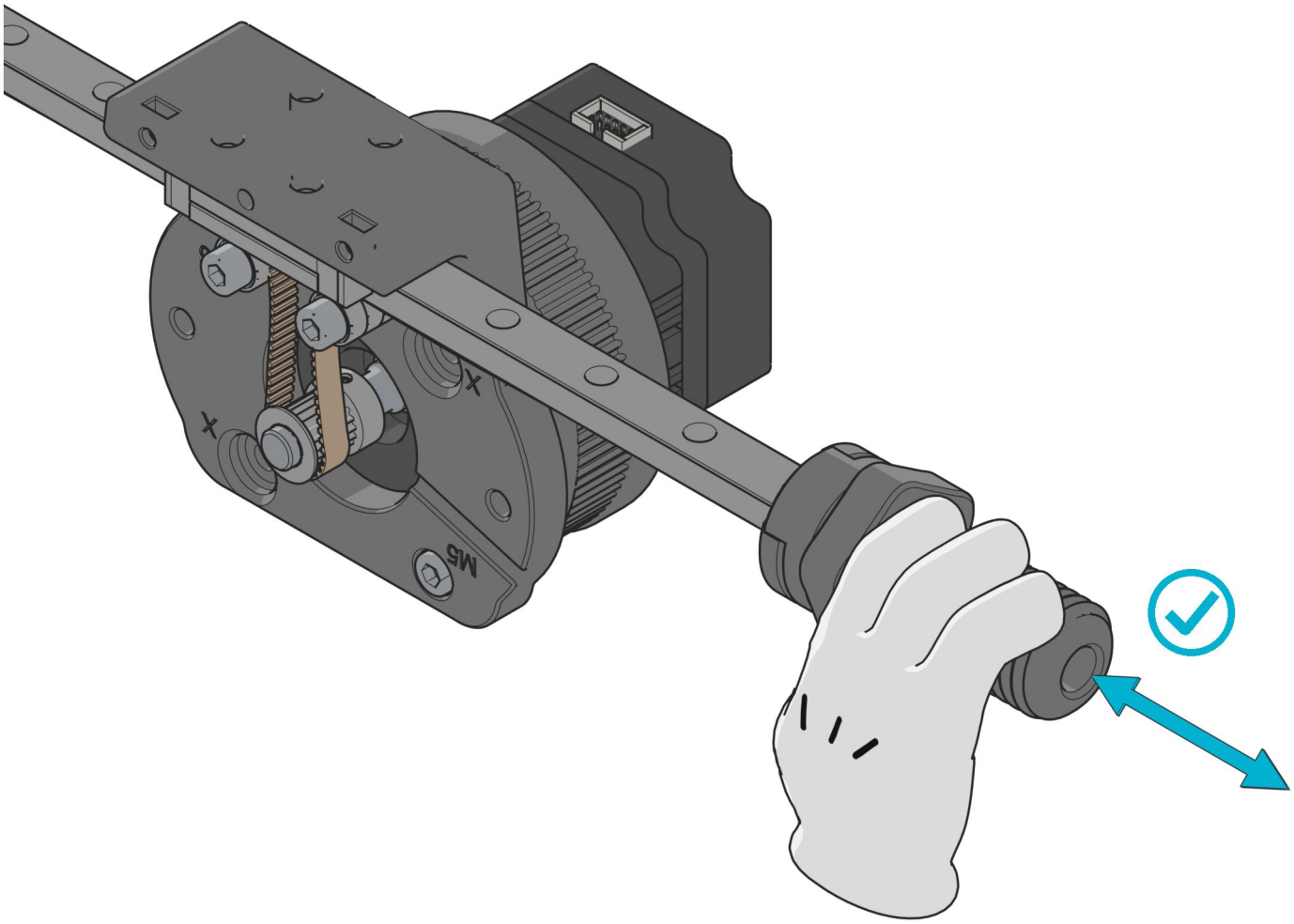
22





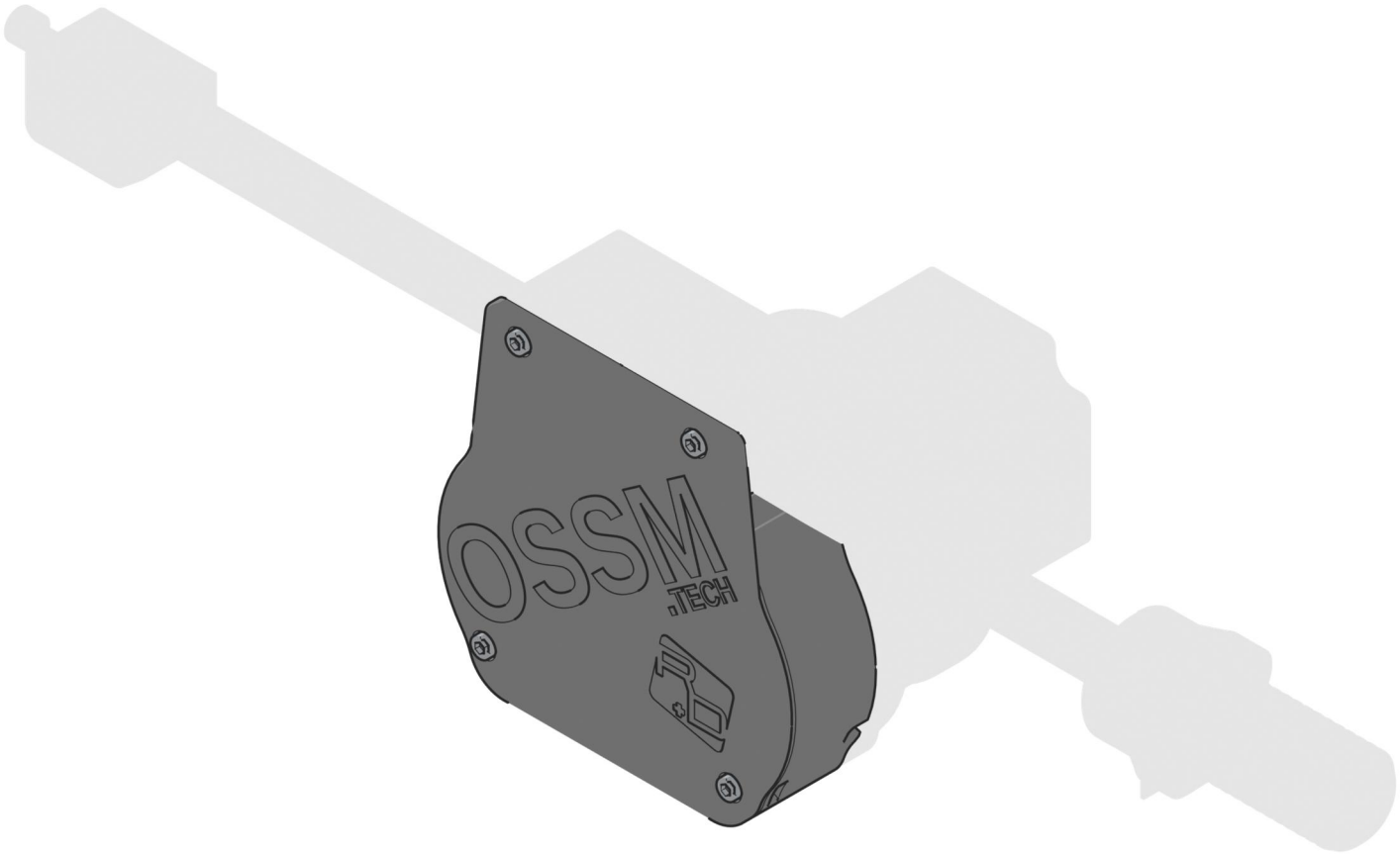
Turn it in by hand: do not over-tighten.

23

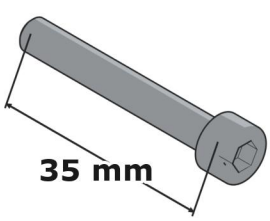


Push and pull the end effector: the rail must slide smoothly through the head.

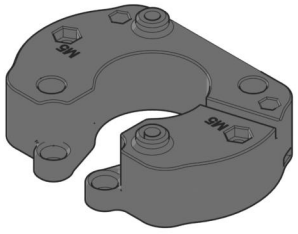
3 Closing the head



24



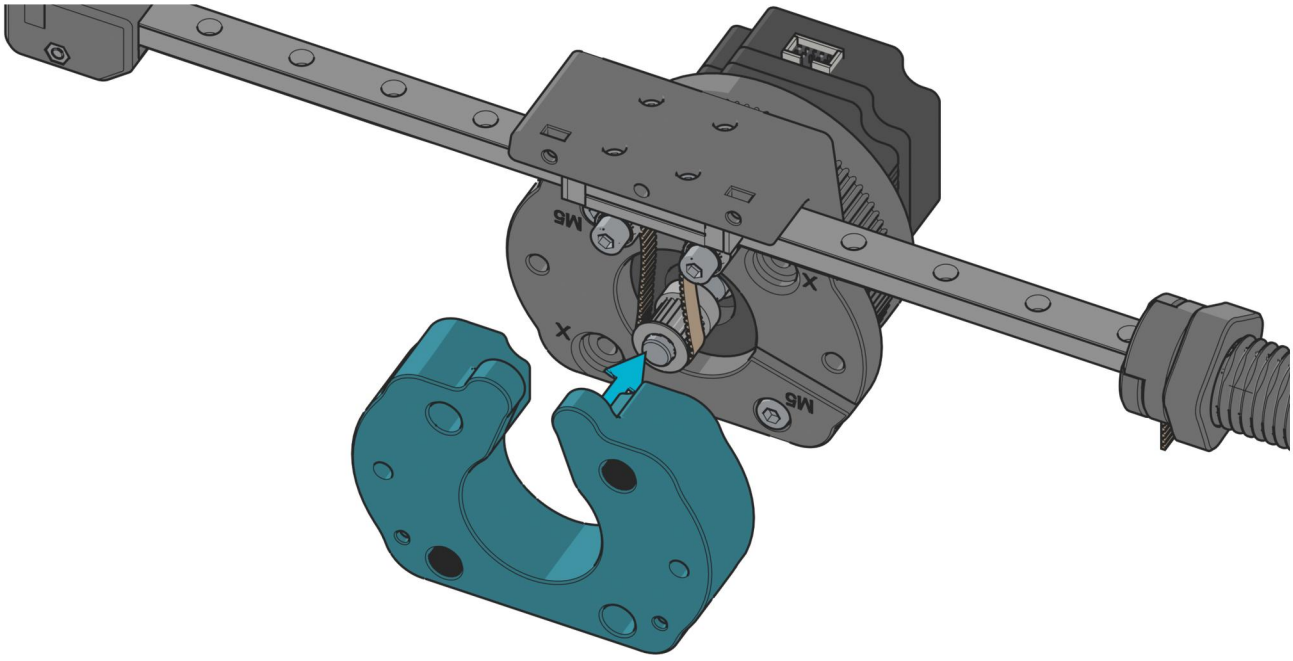
2× M5×35 Screw

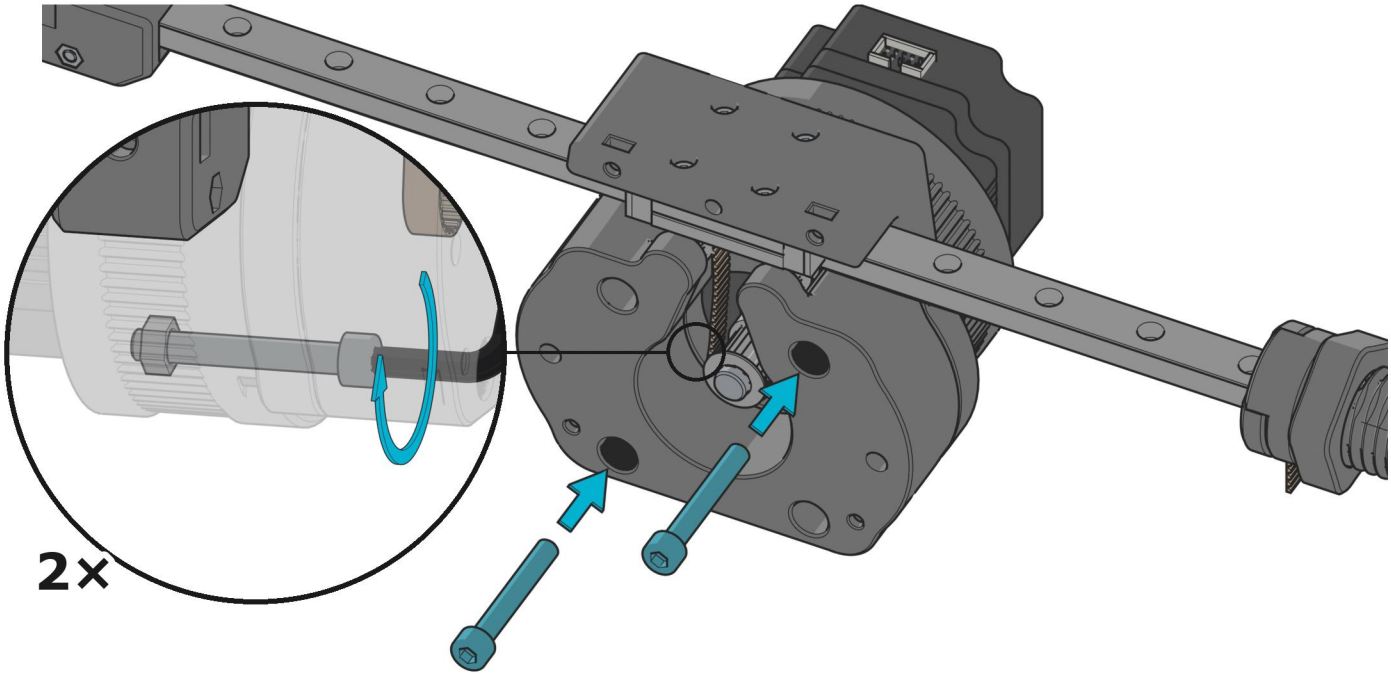


1× Body middle



4 mm Hex key



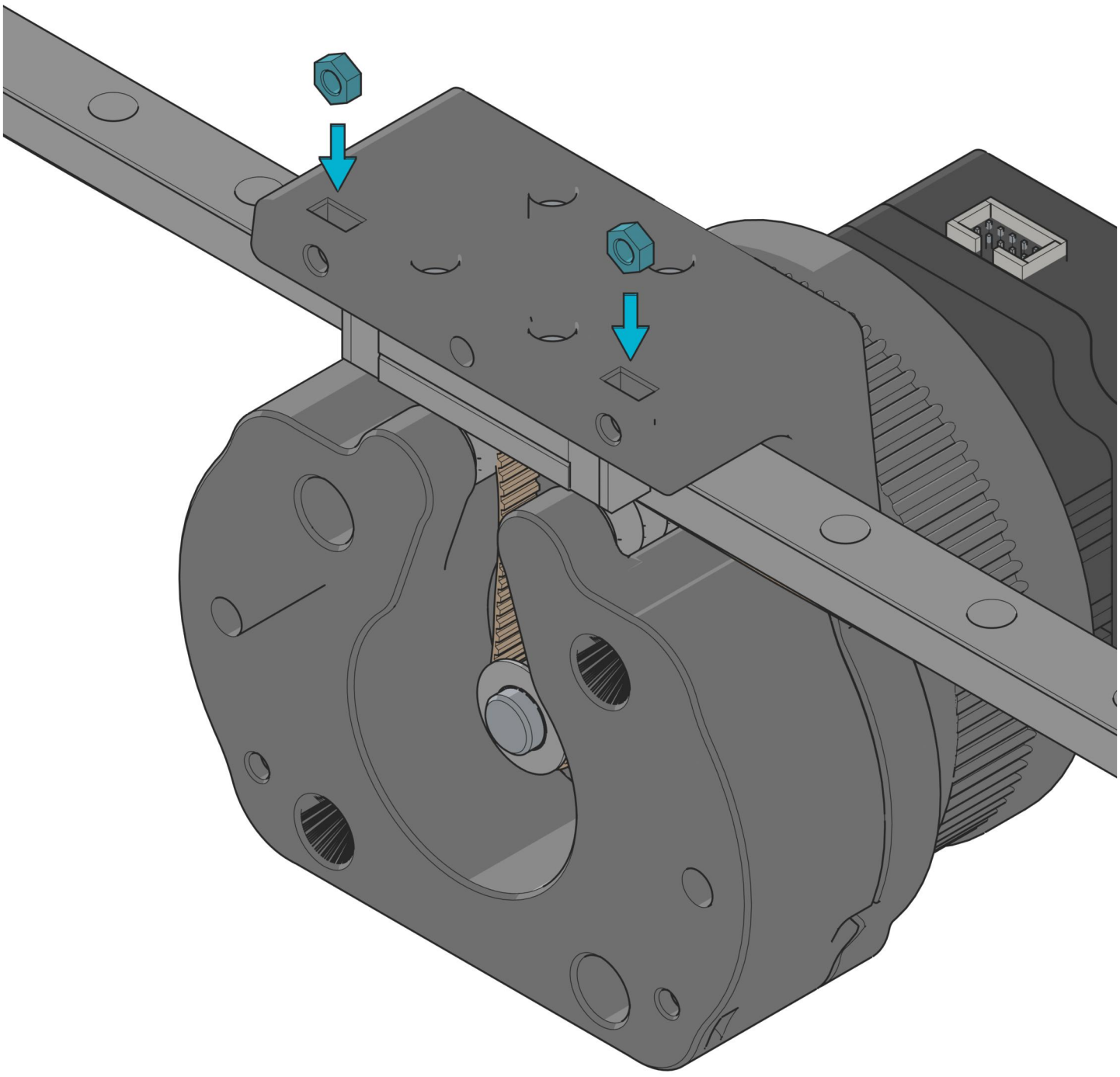


25



2×

M3 Nut

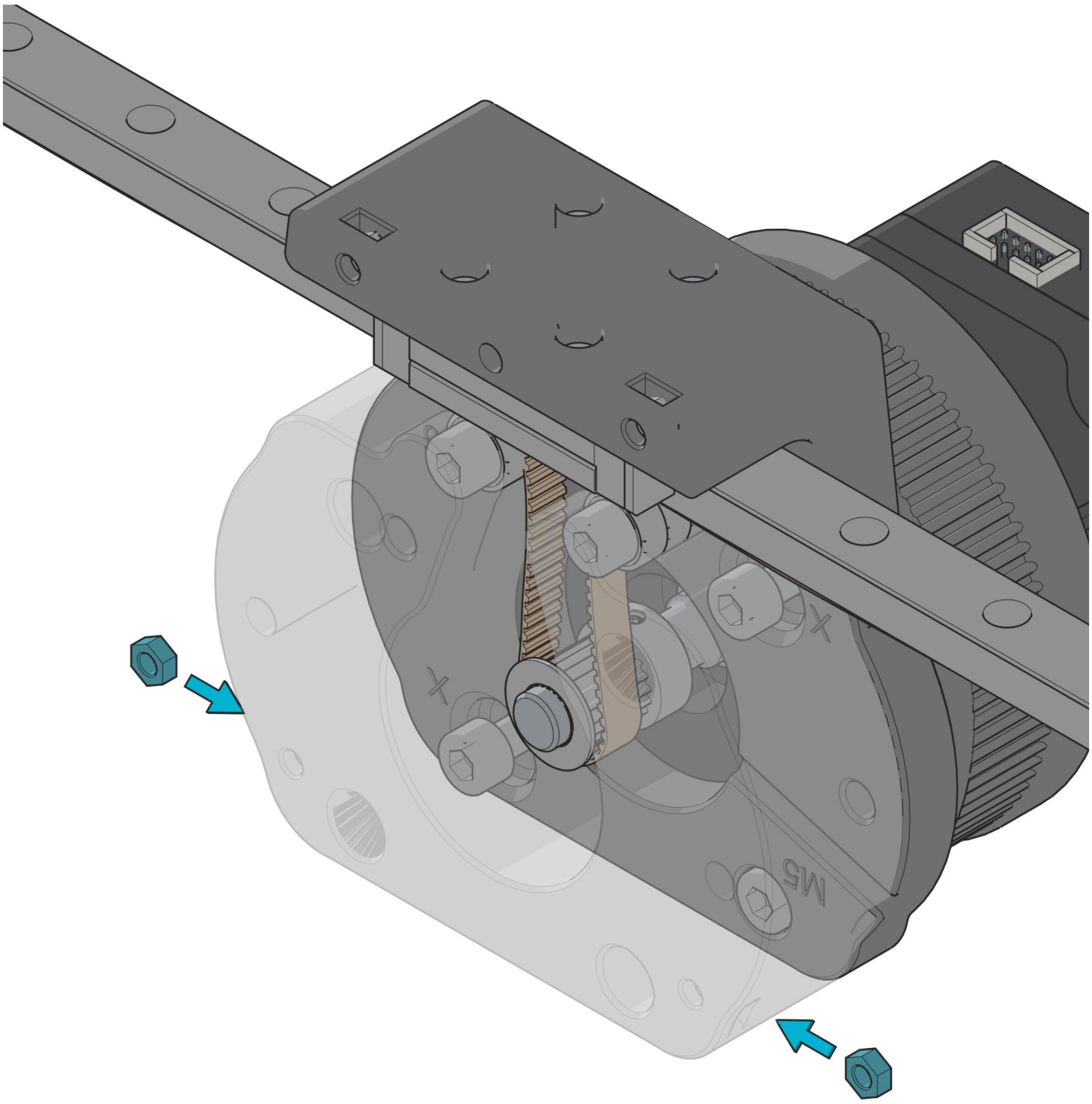


26



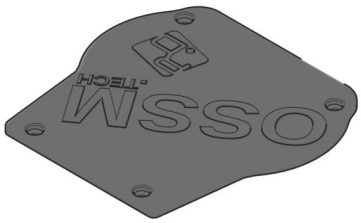
2×

M3 Nut



27



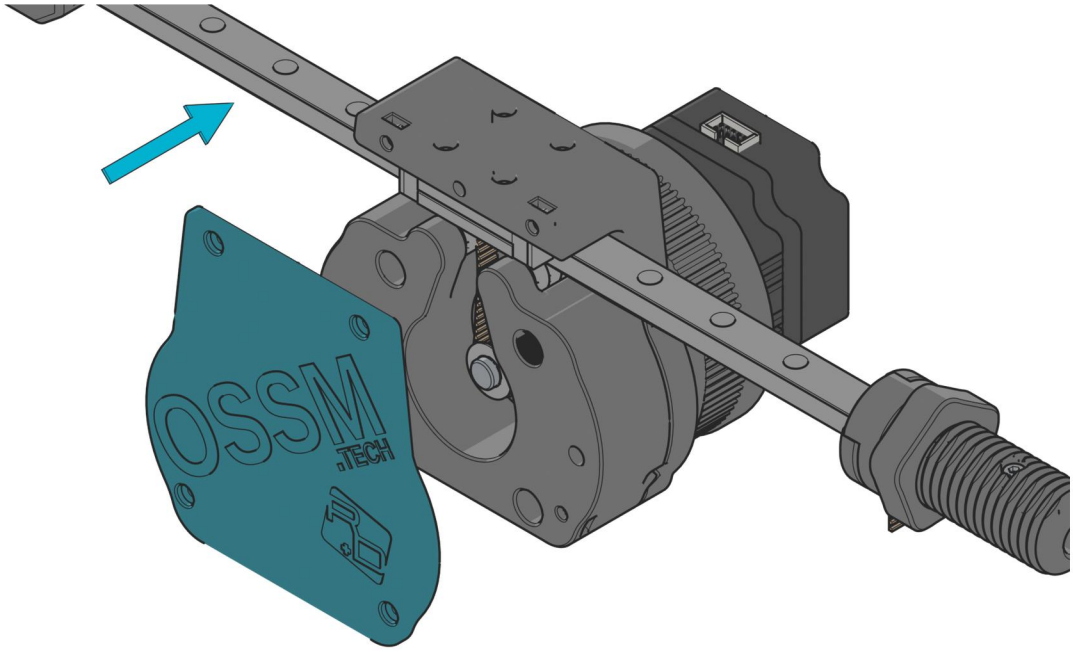


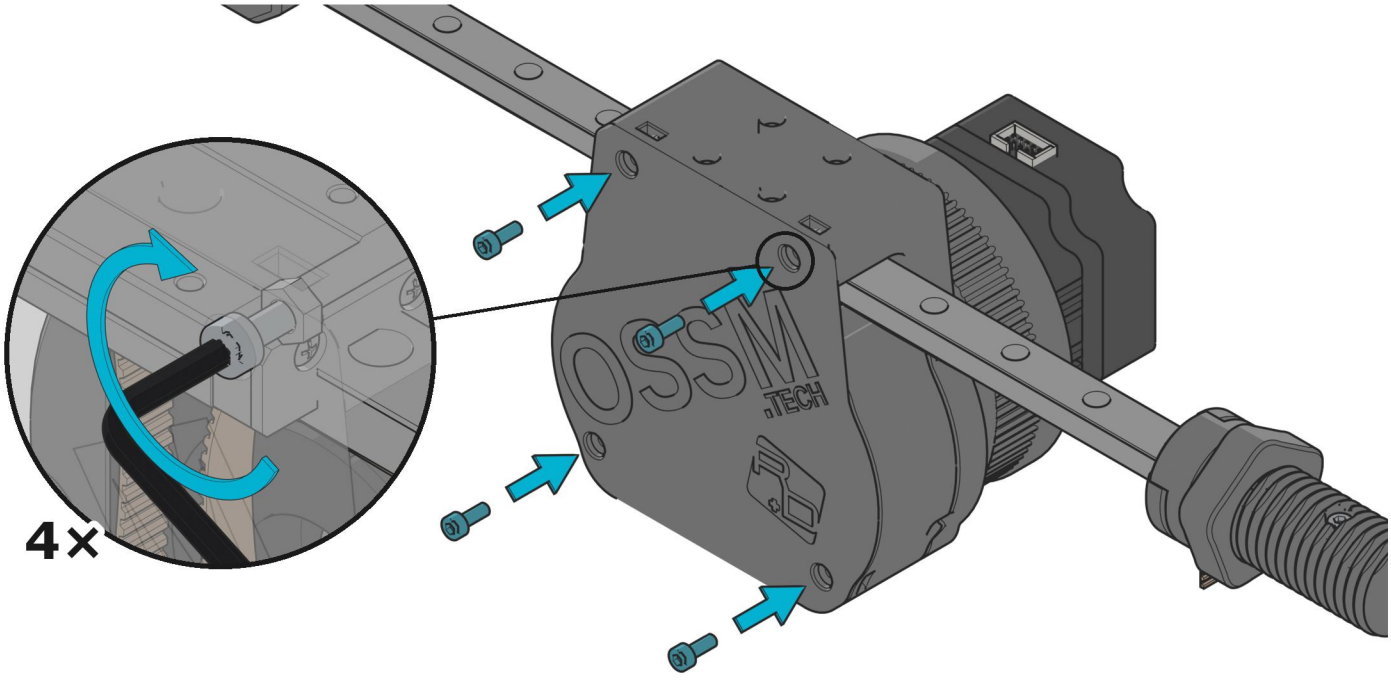
4×
M3×8 Screw

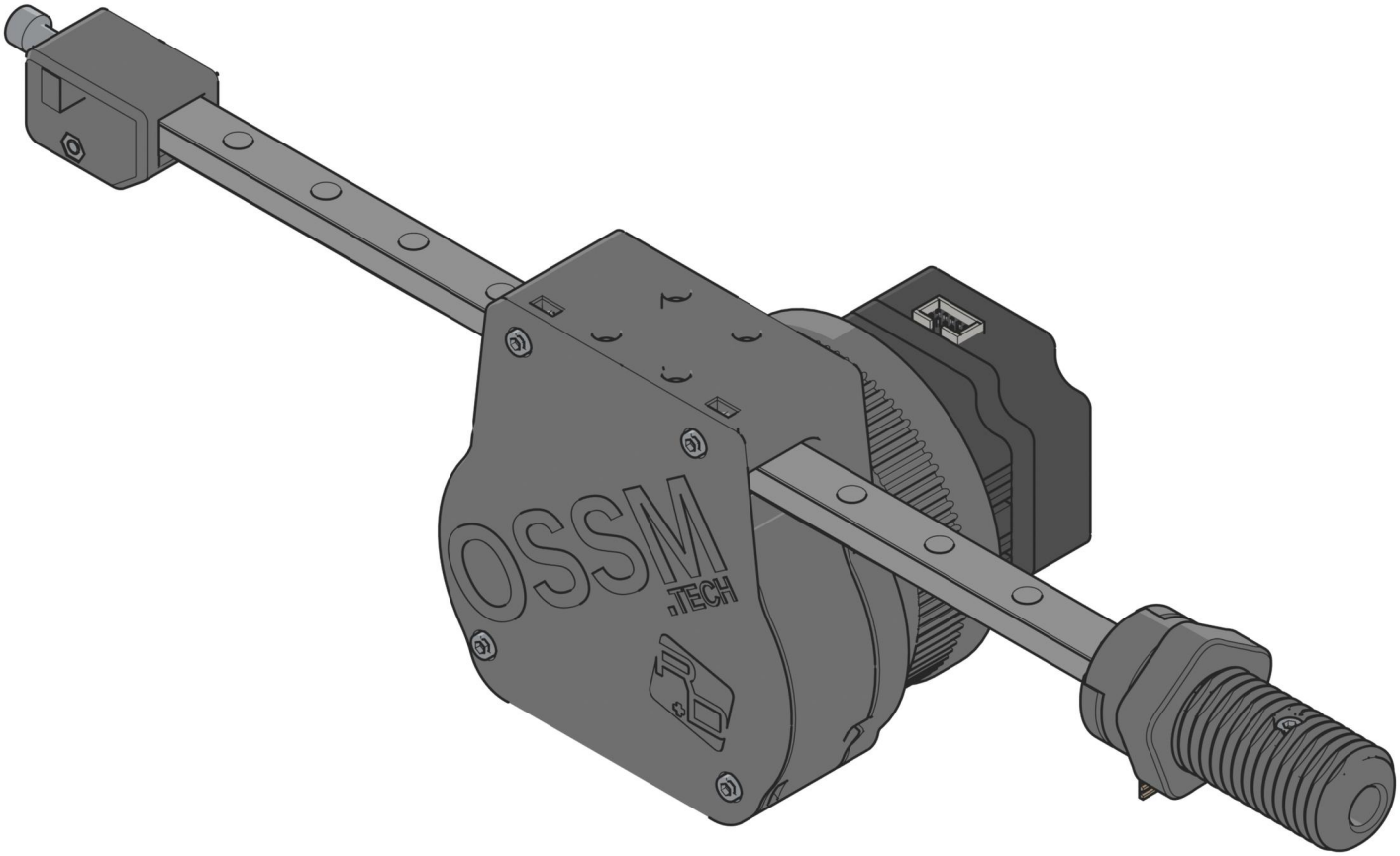
1×
Cover



2.5 mm Hex key







You may now proceed to the electronics and wiring.

RESEARCH + DESIRE

OSSM Actuator Build Guide

Actuator · Version 1.1.13

Support
support@researchanddesire.com
Assembly instructions
ohai.researchanddesire.com/ossm

Gloved hand: "Low Poly Gloved Hand" by CitrusH (@venture.dalton), CC BY 4.0 (creativecommons.org/licenses/by/4.0), posed and restyled: sketchfab.com/3d-models/low-poly-gloved-hand-1f5b60b45c4a43a0984042e23ced7fc6